Collecting Bills and Maintaining the Tax Base: Officer Preferences and Property Records in Punjab

Ahsan Zia Farooqui

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Abstract

A tax authority can collect every bill it issues and still tax some properties too lightly and others too heavily. I study how frontline officers in Punjab, Pakistan, weigh collecting unpaid bills against assessing the full amount due, and how the property records they administer shape the distribution of tax liabilities. In an adaptive choice experiment with 289 tax officers, I estimate that 60.2% prefer a one-percentage-point reduction in delinquency at the cost of a one-percentage-point increase in underassessment, holding total collections and public support fixed. Five years of administrative records from Lahore show how discrepancies in the tax base persist. Recorded land area remained unchanged for 92.3% of 21,816 properties. Among 1,927 unchanged records matched to independent field measurements, 45.0% differed substantially from measured land area. These discrepancies matter for who owes tax. Applying the same statutory schedule to remotely sensed estimates of land and covered area shifts liability towards higher-value properties, raising the Kakwani index of progressivity from 0.042 to 0.090. Mean simulated liability rises by 3.0%, an estimate statistically indistinguishable from zero. A revenue authority that judges performance by collections alone can therefore overlook persistent errors that alter the progressivity of the tax it administers.

1 Introduction

Effective urban service delivery requires a dependable revenue base. As cities expand, water, sanitation, transport and the maintenance of public infrastructure create growing expenditure needs. Raising local revenue gives governments resources to meet those needs and can strengthen the connection between what residents pay and the services they receive. Recurrent property taxation is particularly well suited to this task: it draws on a visible, immobile base whose value often rises with urban growth and public investment (Grote and Wen, 2024; Wani et al., 2020).

Property taxation also offers a way to make the financing of urban services more equitable. Where valuable property is concentrated among wealthier households, a tax that tracks property values can place a larger burden on those holding more wealth. Revenue can then finance services with benefits extending beyond property owners. As land supply is fixed, property taxation provides an efficient way to raise resources without discouraging productive activity. Realising the redistributive potential therefore depends on valuation, the rate schedule, relief and implementation (Blöchliger, 2015). These features make accurate assessment central to both revenue and the distribution of the tax burden.

A good property-tax system must connect each stage of administration. Prichard et al. (2022) organises these stages around identifying properties, valuing them, applying tax policy, billing and securing payment. A complete register, reliable physical attributes, regularly updated valuations, transparent bills and credible enforcement are complementary parts of that system. Procedures for correcting errors and contesting assessments help sustain consistent treatment. This is the tax-systems perspective of Slemrod (1990): the choice of a tax base and rate must be considered together with the information, administrative effort and compliance costs required to implement them. A schedule's intended distribution is realised through the records and decisions used to apply it.

The gap between this potential and practice takes different forms across countries. Property-tax receipts in lower-income countries often remain below 0.2% of GDP (Prichard et al., 2022). Problems also arise where property taxes are established revenue sources. In the United Kingdom, England's Council Tax bands remain tied to April 1991 values (Valuation Office, nd). In the United States, California's Proposition 13 generally limits annual increases in base-year assessed values to 2%, with reassessment principally upon ownership changes or new construction; similar properties can consequently face different bills (California State Board of Equalization, 2025). Germany's former system used 1964 values in the west and 1935 values in the east, producing unequal treatment that prompted constitutional intervention and a new system from January 2025 (Federal Ministry of Finance, Germany, nd). Together, these cases show how valuation rules and their updating can constrain revenue responsiveness and equal treatment even in administrations with substantial capacity.

The obstacles are political as well as technical. Taxing valuable property creates visible liabilities for owners, including groups able to resist revaluation, secure exemptions or constrain rates. Meanwhile, registering properties and valuing them consistently require information that may be incomplete, costly to collect or difficult to verify. Weak transaction data complicate market valuation, while simpler methods depend on the quality of recorded characteristics. Political resistance can thus preserve outdated values, and weak administrative systems can sustain departures from the intended base (Prichard et al., 2022; Grote and Wen, 2024). Reform must address both the rules governing liability and the organisation that translates those rules into bills.

These challenges are especially consequential in Pakistan, where expanding cities face substantial service deficits. Wani et al. (2020) connect Punjab's urban infrastructure needs to the opportunity to mobilise revenue from land and property. In 2020, Punjab collected less urban property tax than Chennai, despite its much larger population. Using the 2021–22 Punjab budget and international comparators, Abbas et al. (2023) report property-tax revenue of about 0.05% of provincial GDP, roughly one-seventh of their lower-middle-income-country benchmark. On the eve of the reform studied here, the Punjab Annual Budget Statement reported revised 2023–24 urban immovable property-tax receipts of PKR 18.36 billion, equivalent to approximately 0.58% of total general revenue receipts of PKR 3,144.58 billion (Government of the Punjab, 2024a).[1]

Successive diagnoses identify persistent weaknesses in Punjab's system. Nabi and Sheikh (2011) emphasise outdated valuation tables, incomplete coverage, exemptions and preferential treatment of owner-occupied property. Wani et al. (2020) explain the trade-offs between valuation accuracy, administrative demands and transparency. Abbas et al. (2023) show how the pre-reform rental-value system combined weak revenue mobilisation with regressive residential burdens, including discounts for larger properties and valuations that poorly captured differences in property values. Their proposed move towards capital-value assessment addresses the valuation rule. Its implementation also depends on the property-specific information to which the new rule is applied.

Frontline officers are central to producing and maintaining that information. Piracha and Moore (2015) describe how Punjab's circle staff control detailed knowledge about properties and arrears, leaving supervisors dependent on information supplied from below. In the present study's setting, inspector-led teams inspect properties and prepare recommended amendments; the Excise and Taxation Officer must approve changes. Informational control and formal approval therefore sit at different levels. Experiments by Khan et al. (2016, 2019) establish that performance pay and performance-based postings can raise collections. The performance-pay evidence also connects reassessment to bargaining over private payments. These experiments identify responses to changes in rewards. The present paper examines how officers rank assessment and recovery outcomes at a given level of receipts.

Punjab's recent reform sharpens this issue. The Punjab Finance Act 2024 replaced annual rental value with capital value as the valuation basis, with implementation from January 2025 (Government of the Punjab, 2024b). The new calculation combines recorded land and covered area with published district-collector rates, making a bill easier to reproduce. The reform carried forward physical inputs from the existing records, so maintaining accurate inputs remained an administrative task. Qualitative interviews for this study describe increasing revenue as a central objective among senior officials, alongside frontline officers' anxiety about higher collection targets.

This is a principal–agent problem with multiple tasks. The revenue authority delegates assessment, record maintenance and collection to officers whose information and effort are difficult to observe. Multitask agency theory explains why the relative rewards attached to these activities matter (Holmström and Milgrom, 1991a). Recovery brings in revenue against an existing bill. An upward correction requires documentation and approval and can create additional demand that must subsequently be collected. These administrative costs and collection obligations can favour recovery even without a private payment. Separately, retaining a lower

[1] Calculated from the detailed UIPT subtotal on printed p. 8 and general revenue receipts on p. 4 of the 2024–25 Annual Budget Statement: $18{,}357.500/3{,}144{,}578.915 \times 100 = 0.584\%$, with amounts in PKR million. Total revenue includes federal receipts; it differs from provincial own tax revenue.

assessment can preserve a taxpayer saving over which the officer and taxpayer may bargain; officers' own views about appropriate burdens can also affect the attraction of correction. The model organises these distinct returns to the two tasks.

The central question is therefore: *Do frontline tax officers value correcting underassessment and recovering unpaid tax differently when both deliver the same collections?* I examine this question by combining an adaptive choice experiment within a survey that holds displayed collections and public support fixed, alongside administrative record histories and independent property measurements. The experiment identifies officers' stated priorities between unresolved assessment and payment problems. The record evidence establishes the maintenance problem to which those priorities are relevant, while independent measurement makes its revenue and distributional consequences visible. This connects the agency problem to a practical requirement of property-tax reform: sustaining the information needed to translate a statutory schedule into accurate liabilities.

Results show inherited discrepancies in the property records used to calculate tax liabilities. In Lahore, 92.3% of 21,816 properties retained the same recorded land area over five years. Among the 1,927 unchanged records linked to field measurements, 45.0% differed substantially from the measured area. An adaptive choice experiment with 289 tax officers identified a tension between collecting existing bills and correcting underassessment. Holding collections and public support fixed, an estimated 60.2% of officers favoured an outcome with less tax left unpaid even when more tax went unassessed. For a one-percentage-point reduction in underassessment accompanied by a one-point increase in delinquency, the median additional revenue needed to offset that trade-off was PKR 4.96 million if collected from low-value properties and PKR 13.85 million if collected from higher-value properties, among officers who positively valued additional collections from the respective group. The records also mattered for who owed tax. Holding the statutory schedule fixed, replacing recorded land and covered areas with remotely sensed estimates raised mean simulated liability by 3.0%, with a 95% interval for the change that included zero, while increasing the Kakwani index of progressivity from 0.0421 to 0.0897. Thus, a modest estimated change in total liability concealed a shift in the distribution of bills towards higher-value properties.

This paper contributes to three strands of research. First, it extends work on incentives in tax administration. Field experiments in Punjab show that financial rewards and performance-based postings can change how much tax officers collect (**?**Khan et al., 2019). I show that officers also care about how a given level of revenue is raised. At equal collections and public support, a majority of officers in the choice experiment favour leaving fewer existing bills unpaid, even when this means more property is taxed below the amount due. Second, the paper contributes to research on information and fiscal capacity (Besley and Persson, 2014; Carrillo et al., 2017) by distinguishing the ability to verify a tax calculation from the ability to verify the records on which it rests. Punjab's reform made bills reproducible from published rates, yet inherited discrepancies persisted, and independent land measurements help identify which properties warrant inspection. Third, the paper contributes to research on the distributional consequences of property tax assessment (Avenancio-León and Howard, 2022). Holding the statutory schedule fixed, I show that replacing recorded property dimensions with independent estimates shifts a larger share of assessed liability towards higher-value properties while producing only a modest estimated change in the average bill. The accuracy of the tax register therefore matters for how the burden is distributed, even when its aggregate fiscal consequences appear small.

The rest of Section 3 describes the institutional setting. Section 4 develops the assessment

and record-correction model. Section 5 describes the data and measurement. Section 6 presents evidence on inherited distortions and persistent discrepancies. Section 7 examines officers' preferences over recovery and underassessment, including compensating public-revenue equivalents. Section 8 examines preferred tax burdens, the concentration of potential tax savings, and the division and acceptance of hypothetical paid concessions. Section 9 evaluates independent measurement, verification targeting and the fiscal and distributional consequences of replacing recorded inputs under a common schedule. Section 10 discusses the implications, and Section 11 concludes. The appendices provide institutional and survey exhibits, valuation and sampling details, model derivations and additional results.

2 Multitask Agency and the Administration of Tax Compliance

Fiscal capacity depends on the organisation of revenue administration. Jensen and Weigel (2026) place personnel, tasks and organisational design at the centre of public finance in developing countries. This paper examines a principal–agent problem within that agenda: an authority delegates both establishing the correct liability and collecting the amount already billed. Its central question is whether officers value these administrative outcomes differently even when collections are the same. Persistent property-record discrepancies give this administrative distinction distributional importance.

Multitask agency theory provides the organising framework. Holmström and Milgrom (1991b) show that incentives affect the allocation of effort across duties, with rewards for measurable output potentially diverting attention from less readily measured quality or maintenance. Rasul and Rogger (2018) document contrasting associations of managerial autonomy and incentive/monitoring practices with project completion in Nigeria. Here the distinction also concerns what a task does to the performance obligation: recovering an existing bill reduces unpaid demand, whereas correcting an understated assessment can create additional demand to recover. The paper measures officers' relative valuations of the unresolved assessment and payment problems through choices holding collections and public support fixed.

The closest tax-administration experiments establish that officers respond to incentives. In Punjab, Khan et al. (2016) examine performance pay, reassessment, accuracy and private bargaining; Khan et al. (2019) show that performance-based postings increase collections. These studies already examine assessment and private returns. The present contribution concerns a different object: officers' preferences over the composition of administrative shortfalls at a given collection outcome. The adaptive choice experiment elicits this trade-off and expresses its strength in compensating public revenue. Administrative histories establish the persistence of the maintenance problem alongside this direct evidence on stated priorities.

Non-collusive compliance channels give this distinction a wider public-finance basis. Brockmeyer et al. (2019) show that credible enforcement emails and information about third-party reports increase filing and payment in Costa Rica. Okunogbe and Pouliquen (2022) show that electronic filing reduces compliance time and changes the pattern of payments across firms in Tajikistan. These experiments estimate how interventions change taxpayer behaviour; this paper elicits how officials rank unresolved tax problems at fixed collections. In Punjab, Piracha and Moore (2015) describe officers' concern with obtaining payment, including lower assessments intended to facilitate collection. Enforcement, information, compliance costs and anticipated payment thus provide channels conceptually separate from sharing a private

surplus.

This distinction motivates the model's separate rewards for assessed liability and receipts, its penalty for uncollected demand, and its amendment cost. In its positive-shortfall comparison at equal additional receipts, the fiscal payoff from upward correction relative to recovery is $(\mu - a)g$: new liability receives credit but also expands the collection obligation. Thus the fiscal component favours recovery when the shortfall penalty exceeds the credit for establishing liability. Amendment work can further discourage correction even with no private payment. The experiment supplies stated valuations of these administrative outcomes; the model explains how non-collusive incentives can generate the trade-off while retaining distinct roles for preferred burdens, career recognition and private returns.

Private deals provide a complementary channel. Prendergast (2007) connects bureaucratic preferences over client outcomes to effort and selective correction, while the Punjab bargaining evidence links private payments to official liabilities. Preferred burdens and monetary gains therefore enter separately: an officer can favour a lower bill without receiving a payment, while a paid concession assigns a share of the saving to the assessor. The chapter's schedule choices and hypothetical concession exercises describe these preferences and potential private stakes alongside the recovery-priority evidence.

Finally, Avenancio-León and Howard (2022) establish distributional consequences of assessment and appeals, and Knebelmann et al. (2026) demonstrate how officer-entered characteristics and valuation knowledge affect accuracy and equity in Senegal. Those studies already connect implementation to unequal burdens. This paper follows inherited physical inputs through record histories, places their persistence alongside elicited administrative priorities, and evaluates the distribution obtained from alternative measurements under a common schedule. Independent measurement also makes assessment quality more verifiable: field validation and discrepancy-based targeting connect the maintenance task to information available for supervision. The broader contribution is to distinguish collecting revenue, establishing liability and maintaining its distributional basis within the same delegated administration.

3 Institutional Setting

3.1 Property-value Assessment and the Role of Front-line Tax Officers

Punjab's Urban Immovable Property Tax (UIPT) is administered by the provincial Excise and Taxation Department. The department divides the urban jurisdictions subject to UIPT into a four-level geographical hierarchy, illustrated in Figure A.1. The primary unit for property inspection, assessment, collection, and enforcement is called the tax "circle". Each circle is assigned an inspector, supported by a clerk and a constable. The inspector leads property visits, determines the relevant property characteristics, and takes primary responsibility for the recommended assessment. The clerk and constable assist with measurement, documentation, record preparation, and the delivery of notices. Accordingly, the term "officer" in the theoretical analysis refers to the inspector as the relevant front-line decision-maker.

The circle team cannot enter its recommendation directly into the official tax record. It must submit the assessment to the Excise and Taxation Officer (ETO), whose approval is mandatory. The ETO reviews the measurements, classifications, and supporting record prepared by the inspector-led team. The process therefore separates informational control from formal authority: the inspector controls the production of property-specific information, while the ETO controls

whether the recommendation receives formal approval. For analytical tractability, the model assumes that the ETO is not a party to any informal bargain between the inspector and the taxpayer. ETO approval nevertheless matters because the inspector must produce a record that is sufficiently complete and internally consistent to obtain formal approval.

3.2 The Capital-value Reform

Before 2025, UIPT liability was based on annual rental value (ARV). The inspector recommended a rental value by applying the department's notified valuation tables, reproduced in Figure A.2, to the property's location, use, occupancy, condition, and physical characteristics. Although the resulting rental value appeared in the administrative record and informed the taxpayer's bill, the taxpayer could not independently determine the correct rental value from publicly observable information. The tables constrained assessments but left substantial room for judgement about prevailing rents and property characteristics (Piracha and Moore, 2015; **?**). Consequently, a low assessment could reflect either a legitimate valuation judgement or an officer-mediated departure, and the administrative record alone could not distinguish between them.

Figure A.3 shows the tax schedule under the Punjab Finance Act 2024 Amendment, which replaced annual rental value (ARV) with capital value (CV) as the statutory valuation basis. The change took effect on 1 January 2025, during fiscal year 2024–25. Under CV, recorded land and covered area are valued using the applicable district-collector neighbourhood rates, referred to below as DC rates. The resulting capital value is taxed under the value-band schedule illustrated in Figures A.4.

3.3 Verifiability and the Inherited Cadastre

The reform changed both the valuation basis and the information required to support an assessment. Under ARV, the recorded value depended on judgements that a taxpayer or supervising officer could not reproduce independently. Under CV, the statutory calculation combines recorded physical attributes with published rates. The relevant property characteristics and CV details appear in the administrative record and on the converted taxpayer's bill. A taxpayer, supervisor, or other third party can therefore reproduce the official calculation and determine whether it is internally consistent with the recorded inputs. Figure A.5 illustrates this change in the information displayed on the bill.

The reform did not, however, introduce a general remeasurement of properties. The land area and covered area used under CV were inherited from the ARV-era cadastre. It also left the audit budget, inspection resources, officer remuneration and benefits, sanction schedule, frequency of review, and disciplinary chain substantially unchanged. Qualitative evidence from officials confirms that the principal institutional change concerned the valuation rule rather than enforcement intensity or officer incentives outside the assessment process.

The distinction between verifiability and accuracy is therefore central. A CV assessment can be reproduced exactly from the official record even when the recorded land or covered area is stale or incorrect. Greater verifiability may induce officers to correct some records because inconsistencies are easier for taxpayers and supervisors to detect, but the formula does not mechanically correct its own inputs. Field measurements are then used to determine whether remote sensing provides a more accurate benchmark than the administrative record.

3.4 Recovery and correction

Demand is the total amount billed for collection. The collection-target framework treats recorded demand as the obligation against which recovery is assessed; the exact treatment of each amendment requires institutional verification. *Recovery* means obtaining payment against that demand. *Correction* means amending an inaccurate property entry, which can increase, decrease or leave unchanged the bill calculated from it. Underassessment is a bill below the relevant benchmark; delinquency is nonpayment of an amount already assessed. These are different gaps in the tax system.

Recovery improves collections against an existing obligation. An upward correction can expand the obligation. The officer must first verify and document the change and secure approval; if the higher bill is not fully paid, the circle also has a larger amount left to recover. Officer accounts in the project's institutional material describe field verification, approval requirements, collection pressure and carried-forward records. They motivate these administrative costs without implying that every unchanged entry is inaccurate or that every officer resists correction.

4 Theoretical Framework

A tax bill can follow the prescribed calculation and still be wrong because its inputs are inaccurate. I distinguish two decisions. First, an inspector can assess below the amount implied by an existing record. Second, the inspector can keep an inaccurate record instead of correcting it. Either decision can leave the taxpayer with a lower bill and provide an opportunity to negotiate a bribe. I use the same Nash bargaining framework for both negotiations, with different payoffs and different outcomes if agreement fails.

In each negotiation, the taxpayer compares the tax saving with the bribe. The inspector compares the bribe with the administrative benefits and costs of the alternative. Both value a rupee received at one payoff unit. Bargaining weights describe how they divide the gain from agreement; the bribe and its share of the tax saving are determined within the model. I first examine assessment holding the record fixed, then record correction holding the calculation rule fixed. These are separate comparisons, rather than a joint solution for simultaneous changes in the record and the calculation.

4.1 Assessment conditional on the record

For property i, let $V_i^{\dagger} \geq 0$ be the property value assessment implied by correct property attributes. Let $\bar{V}_i$ be the property value before any departures caused by officers' assessment, such that $\bar{V}_i = V_i^{\dagger} + \xi_i$, where ξ is measurement error during property assessment. The inspector further chooses a downward departure $d_i \in [0, \bar{V}_i]$, producing $V_i = \bar{V}_i - d_i$. Before record amendment, $\bar{V}_i = V_i^0$, the inherited-record formula value defined in Section 4.2. Let $T_i(V_i)$ be annual tax liability at assessment V.

Assumption 1. Throughout this paper, I assume that the officer chooses to underassess from the record-based benchmark such that $d_i \geq 0$ and $V_i \leq \bar{V}_i$.

Let $r \in \{\text{ARV}, \text{CV}\}$ denote the valuation regime. The regime-specific cost of a checkable departure satisfies $0 \leq \gamma_{\text{ARV}} < \gamma_{\text{CV}}$, reflecting greater supervisor verifiability under capital-

value taxation. Suppressing the inspector index, the original payoff is

$$\begin{aligned}\Pi_i(d_i) = \mu T_i(\bar{V}_i - d_i) + \theta\{T_i(\bar{V}_i) - T_i(\bar{V}_i - d_i)\} \\ - \gamma_r d_i^2 - \delta p_i(d_i).\end{aligned} \tag{4.1}$$

The weight $\mu \in [0, 1)$ values assessed liability credited to the revenue objective.

The coefficient $\gamma \geq 0$ measures the cost of making a checkable departure, $\delta \geq 0$ is the loss if it is scrutinised, and $p_i(d_i)$ is the probability of scrutiny. Supervisor scrutiny occurs with probability $p_i(d_i) = \bar{p}(d_i/\bar{A}_i)^2$, where $\bar{p} \in [0, 1]$.

Maintained assumptions. The tax schedule is finite, nonnegative and nondecreasing, and the officer's maximum is attained. The verifiability comparison holds the record, benchmark, tax schedule, θ, μ, δ, and $\bar{p}$ fixed. The FOC below applies to a positive interior optimum within an affine tax segment of marginal rate $\tau > 0$; it does not impose a globally proportional tax schedule. Payoffs are measured in monetary-equivalent units for the assessment cycle. The revenue term captures the administrative value of contributing to the department's fiscal objective, reflecting the hierarchical pressure to raise revenue documented by Piracha and Moore (2015).

The negotiated bribe. Underassessment costs the inspector some assessed-liability credit and exposes the departure to scrutiny. Relative to applying the formula without a bribe, the total loss before receiving payment is

$$K_i^a(d_i) = \mu G_i^a(d_i) + H_i d_i^2.$$

The inspector therefore needs at least K_i^a to accept the arrangement. The taxpayer will pay at most G_i^a. An agreement can benefit both when $G_i^a > K_i^a$. Let $\beta_a^B \in (0, 1)$ be the inspector's bargaining weight. Nash bargaining gives the inspector compensation for the loss K_i^a, plus the fraction β_a^B of the gain left to divide:

$$\begin{aligned}b_i^{a*}(d_i) &= K_i^a(d_i) + \beta_a^B\{G_i^a(d_i) - K_i^a(d_i)\}, \\ \theta_i^a(d_i) \equiv \frac{b_i^{a*}(d_i)}{G_i^a(d_i)} &= \beta_a^B + (1 - \beta_a^B)\frac{K_i^a(d_i)}{G_i^a(d_i)}.\end{aligned} \tag{4.2}$$

The share θ_i^a is an outcome of negotiation, not a fixed parameter. Even with unchanged bargaining power, a more costly departure requires a larger payment for a given saving. Appendix C derives the solution.

The agreed departure. After negotiating the payment, the inspector gains $\beta_a^B[G_i^a(d_i) - K_i^a(d_i)]$ relative to the benchmark. The taxpayer receives the remaining fraction. Both therefore prefer the departure that maximises their combined gain,

$$S_i^a(d_i) = (1 - \mu)G_i^a(d_i) - H_i d_i^2.$$

If no departure gives a strictly positive gain, they make no agreement and $d_i = b_i^a = 0$. The tax schedule is finite, nonnegative and nondecreasing, and the maximum is assumed to be attained. Within an affine tax segment with marginal rate $\tau > 0$, a positive interior optimum satisfies

$$d_i^* = \frac{(1 - \mu)\tau}{2H_i}. \tag{4.3}$$

Assume a finite, nonnegative, nondecreasing schedule and an attained maximum. When $\bar{V}_i = 0$, only $d_i = 0$ is available and $p_i(0) \equiv 0$ is defined separately, avoiding division by zero. Appendix B treats boundaries and proves that. Since $\bar{V}_i = V_i^\dagger + \xi$ and $V_i = \bar{V}_i - d^\star$, V_i can be expressed as $V_i = V_i^\dagger + \xi - d^\star$. Holding the record and other primitives fixed, increasing γ weakly reduces departures and takes d_i^* towards zero. However, the assessment will approaches $\bar{V}_i$, the benchmark of the *existing record*. It need not approach the assessment based on correct inputs $V_i^\dagger$.

4.2 Two records and the reasons to retain or correct them

Let $X_i^t = (a_i^t, s_i^t)$ denote the land and covered-area attributes of property i and where $t \in \{0, \dagger\}$ where 0 denote inherited property attributes and † denotes correct property attributes, and let S_l denote the public rate schedule in locality l. If X_i^0 denotes inherited land and covered-area entries and $X_i^\dagger$ their correct current value, capital value can be expressed as

$$V_i^t = h(S_l, X_i^t). \tag{4.4}$$

The two assessments are $V_i^0 = h(X_i^0)$ and $V_i^\dagger = h(X_i^\dagger)$. They are formula-based assessments, not necessarily market values. Define the two annual bills and their difference:

$$T_i^0 = T_i(V_i^0), \qquad T_i^1 = T_i(V_i^\dagger), \qquad g_i = T_i^1 - T_i^0. \tag{4.5}$$

Thus g_i is positive when correction raises liability, negative when it lowers liability, and zero when both records imply the same bill. The superscripts identify the records, not powers. These amounts concern assessed liability, not collections.

Inspector j chooses retention, $u_{ij} = 0$, or correction, $u_{ij} = 1$. The comparison starts after the discrepancy and correct replacement inputs are known to the inspector. Correction completes the authorised amendment; if it is infeasible, the record remains unchanged. Feasibility includes documentary requirements and senior approval, rather than unilateral inspector authority. Conditional on feasibility, completion enters the correct attributes. The inspector knows the costs and payoff parameters below. Calculation is mechanical under either record, so $d = 0$.

A second negotiation. When $g_i > 0$, keeping the old record preserves a lower bill. The inspector and taxpayer may negotiate a bribe b_{ij}^x for retaining it; the superscript x identifies the record margin. The private-return terms are

$$R_{ij}^x(0) = b_{ij}^x, \qquad R_{ij}^x(1) = 0. \tag{4.6}$$

The payment is determined below by comparing retention with correction. If no payment agreement is reached, the inspector chooses whichever action gives the higher payoff without a bribe, retaining the record at a tie. This makes the consequence of refusing a bribe explicit. An inspector who already prefers unpaid retention cannot demand payment by threatening a correction that the inspector would not carry out.

The two negotiations concern different savings. Underassessment creates a saving below the formula value of a given record; retention preserves a saving by keeping that record unchanged. At $d = 0$, the first saving disappears while the second can remain. Both use Nash bargaining, but the administrative alternative and hence the negotiated payment differ. Payments for processing a downward correction are outside these two margins. For $g_i \leq 0$, or an infeasible correction, I set $b_{ij}^x = 0$.

Preferred bills and amendment costs. Let $P_{ij} \geq 0$ be the annual bill inspector j thinks the property should pay. A difference from that amount reduces the inspector's payoff by $\lambda_j |T_i^u - P_{ij}|$, with $\lambda_j \geq 0$. The preferred bill is the same under both actions. Correction also requires remaining effort and opportunity cost $c_{ij} \geq 0$, including documentation, processing and approval. Discovery costs already incurred are sunk. The preferred-bill gain from correction is

$$q_{ij} = |T_i^0 - P_{ij}| - |T_i^1 - P_{ij}|. \tag{4.7}$$

It is positive when correction moves the bill closer to the inspector's preferred amount. Its contribution to the payoff change is $\lambda_j q_{ij}$.

4.3 Correction when collection obligations matter

Let B_j^u be the circle's booked tax demand under action u and R_j^u the receipts expected against that demand in the same evaluation period. Both are rupee amounts. The shortfall $[B_j^u - R_j^u]_+ = \max\{B_j^u - R_j^u, 0\}$ is expected uncollected demand. The focal amendment changes demand by $B_j^1 - B_j^0 = g_i$ and expected receipts by $r_{ij} = R_j^1 - R_j^0$. Capital R denotes the circle's receipt level; lowercase r denotes the change caused by the amendment.

The inspector values assessed liability with weight μ_j, receipts with weight $\eta_j \geq 0$, and bears a penalty $a_j \geq 0$ per rupee left uncollected. Correction changes career recognition by h_{ij}, valued with weight $\kappa_j \geq 0$. At $d = 0$, equation (4.1) contributes only $\mu_j T_i^u$: there is no departure saving, departure bribe or departure cost. Adding collection incentives and the record-choice terms gives

$$\begin{aligned} J_{ij}(u; b_{ij}^x) = \mu_j T_i^u + \eta_j R_j^u - a_j [B_j^u - R_j^u]_+ \\ + \kappa_j u h_{ij} + (1-u) b_{ij}^x - \lambda_j |T_i^u - P_{ij}| - u c_{ij}. \end{aligned} \tag{4.8}$$

The factor $(1 - u)$ applies the bribe only to retention; correction earns recognition and incurs amendment cost. All terms use the inspector's information and the same monetary-equivalent units.

What correction is worth without a bribe. Define $F_{ij} = J_{ij}(1; 0) - J_{ij}(0; 0)$. This is the officer's gain from correcting rather than retaining the record when neither action brings a private payment. When both actions leave positive uncollected demand, subtraction gives

$$\begin{aligned} F_{ij} &= \mu_j g_i + (\eta_j + a_j) r_{ij} - a_j g_i + \kappa_j h_{ij} + \lambda_j q_{ij} - c_{ij}, \\ \Delta_{ij}(b_{ij}^x) &\equiv J_{ij}(1; 0) - J_{ij}(0; b_{ij}^x) = F_{ij} - b_{ij}^x. \end{aligned} \tag{4.9}$$

Correction earns credit for the additional liability and receipts, but an added bill that is not collected increases the shortfall. Career recognition and closer alignment with the preferred bill can add to the gain; amendment work reduces it. If either action eliminates the shortfall, F_{ij} is calculated directly from the positive-part payoffs in equation (4.8). If booked demand changes only later, its contemporaneous change replaces g_i in the shortfall calculation.

Negotiating to keep the old record. For a feasible upward correction with $F_{ij} > 0$, the inspector would correct if negotiation failed. Keeping the old record saves the taxpayer g_i, while the inspector gives up a correction worth F_{ij}. The taxpayer gains $g_i - b_{ij}^x$ from agreement and

the inspector gains $b^x_{ij} - F_{ij}$. Here the taxpayer values the reduction in assessed liability rupee for rupee; the inspector's collection forecast enters separately through r_{ij}.

When $0 < F_{ij} < g_i$, the saving is large enough to compensate the inspector and leave a gain for both. With inspector bargaining weight $\beta^B_{x,j} \in (0, 1)$, the negotiated payment and its share are

$$\begin{aligned} b^{x*}_{ij} &= F_{ij} + \beta^B_{x,j}(g_i - F_{ij}), \\ \theta^x_{ij} \equiv \frac{b^{x*}_{ij}}{g_i} &= \beta^B_{x,j} + (1 - \beta^B_{x,j})\frac{F_{ij}}{g_i}. \end{aligned} \tag{4.10}$$

The bribe first compensates the inspector for giving up correction, then gives the inspector a share of the remaining gain. A better correction option raises the negotiated payment, even with unchanged bargaining power. The effective share θ^x_{ij} therefore varies across properties and administrative conditions.

Within a paid agreement, $\Delta_{ij}(b^{x*}_{ij}) = \beta^B_{x,j}(F_{ij} - g_i) < 0$: the inspector prefers the agreed bribe to correction. If $F_{ij} \geq g_i$, no payment can make both parties strictly better off than correction. If $F_{ij} \leq 0$, the inspector already prefers unpaid retention, using the stated tie rule. I select no agreement when its joint gain is zero. The resulting correction rule is

$$u^*_{ij} = \begin{cases} 1, & \text{correction feasible, } g_i > 0 \text{ and } F_{ij} \geq g_i, \\ 1, & \text{correction feasible, } g_i \leq 0 \text{ and } F_{ij} > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{4.11}$$

Paid retention occurs only for a feasible correction with $0 < F_{ij} < g_i$; otherwise the equilibrium retention payment is zero. Appendix C.3 gives the participation conditions. Bargaining power affects the division of a feasible agreement, while the comparison of F_{ij} with g_i determines whether an agreement can prevent an upward correction.

Correction versus recovery at equal receipts. Consider two tasks starting from the same demand and receipts. Recovery collects $\Delta R > 0$ without changing the bill. An upward correction adds demand $g_i > 0$ and generates the same receipts, $r_{ij} = \Delta R$. Assume $0 < \Delta R \leq g_i$ and $\Delta R < B^0_j - R^0_j$, so both leave positive uncollected demand. Recovery's fiscal gain is $(\eta_j + a_j)\Delta R$; correction's is $\mu_j g_i + (\eta_j + a_j)\Delta R - a_j g_i$. Their difference is

$$\text{Fiscal payoff from correction minus recovery} = (\mu_j - a_j)g_i. \tag{4.12}$$

The fiscal component favours recovery when the shortfall penalty a_j exceeds the assessed-liability credit μ_j. Amendment costs, career recognition and preferred-bill alignment also matter for the full task comparison. If recovery is the alternative to correction in a retention negotiation, these task payoffs must determine the negotiated payment as well. Appendix C.3 states that comparison. Equation (4.12) isolates the fiscal distinction measured most closely by the officer experiment.

For an upward correction, write $r_{ij} = p_{ij}g_i$, where $p_{ij} \in [0, 1]$ is the expected collection share of the additional bill. For example, an added bill of 100 with $p_{ij} = 0.8$ yields expected receipts of 80 and adds 20 to the shortfall. This share differs from the scrutiny probability $p_i(d_i)$. Substitution into the negotiated correction rule gives

$$\begin{gathered} [\mu_j + \eta_j p_{ij} - a_j(1 - p_{ij})]g_i + \kappa_j h_{ij} + \lambda_j q_{ij} - c_{ij} \geq g_i, \\ g_i > 0, \qquad \text{correction feasible.} \end{gathered} \tag{4.13}$$

Better collection prospects raise F_{ij} and the payment needed to retain the record, until retention can no longer leave a gain for both parties. For a downward correction with receipts unchanged, the fiscal gain is $(a_j - \mu_j)|g_i|$: lower demand reduces the shortfall but also reduces assessed-liability credit. Its decision rule remains $F_{ij} > 0$, since no retention bribe is supported by a saving in that case. Appendix B.1 examines the special case without collection or career terms.

Independent verification of the physical record. Let $K_{ij}^{\text{audit}} \geq 0$ be the expected sanction loss from retaining a known inaccurate record, eliminated by completing correction. This record-audit risk is distinct from scrutiny of a departure below the formula. Its effect on the payoff before a bribe is

$$\begin{aligned} J_{ij}^{\text{audit}}(u; b) &= J_{ij}(u; b) - (1 - u)K_{ij}^{\text{audit}}, \\ F_{ij}^{\text{audit}} &= F_{ij} + K_{ij}^{\text{audit}}. \end{aligned} \tag{4.14}$$

The payment is renegotiated using F_{ij}^{audit}. Within a paid agreement, an additional rupee of expected sanction raises the bribe by $1 - \beta_{x,j}^{B}$ and reduces the gain from retention. An upward correction occurs once $F_{ij}^{\text{audit}} \geq g_i$. The baseline sets this additional loss to zero. Independent measurements can raise expected sanctions if used in supervision, or reduce c_{ij} by making the remaining amendment work easier. Section 9 examines measurement accuracy and discrepancy-based targeting; it does not estimate these responses in sanctions, costs or correction decisions.

4.4 Empirical implications: private returns and recovery priorities

The model connects private stakes in lower bills with officers' preferences over assessment, collection and tax burdens. The empirical exercises examine these components separately; they do not estimate the property-level correction rule.

Private returns and preferred official burdens. Within a paid retention agreement, the negotiated payment responds to the tax saving and the correction option:

$$\left.\frac{\partial b_{ij}^{x*}}{\partial g_i}\right|_{F_{ij}} = \beta_{x,j}^{B} > 0, \qquad \left.\frac{\partial b_{ij}^{x*}}{\partial F_{ij}}\right|_{g_i} = 1 - \beta_{x,j}^{B} > 0.$$

A larger saving permits a larger payment, holding the correction payoff fixed. A more attractive correction option also raises the payment, holding the saving fixed. If both change, both contributions must be included. Stronger officer bargaining power raises the bribe by $g_i - F_{ij}$ per unit increase in the bargaining weight, but does not change the agreement boundary $F_{ij} = g_i$.

The preferred-bill motive affects that correction option. When correction moves the bill closer to what the inspector considers appropriate, $q_{ij} > 0$, the term $\lambda_j q_{ij}$ raises F_{ij}. Within a paid agreement, this raises the compensation required to retain the lower bill; if $F_{ij} \geq g_i$, no mutually beneficial retention payment is available. Conversely, a correction that moves the bill further from the preferred amount has $q_{ij} < 0$ and lowers the correction payoff. The preferred-schedule exercise describes the burdens officers favour, corresponding to P_{ij}. The citizen conjoint describes acceptance and the division of selected hypothetical concessions. Their payment shares do not by themselves reveal bargaining weights or the value of the officer's alternative.

Distributional stakes. The distribution of potential tax savings locates the amounts that could be divided in a retention agreement. For a given correction payoff, larger savings raise the possible private payment. The empirical comparison therefore examines where formula-implied savings are concentrated and how replacing recorded inputs redistributes liabilities under the same schedule. It keeps the size of the saving, its conditional division and acceptance as distinct objects.

Recovery priority and compensating revenue. At equal expected receipts, equation (4.12) identifies the fiscal reason to favour recovery over establishing omitted liability. BACE holds displayed collections and public support fixed while exchanging less delinquency for more underassessment. Its contrast $\omega_j = \beta_{Uj} - \beta_{Dj}$ measures the relative valuation of those outcomes. A positive contrast is consistent with a recovery priority, but does not separately identify $a_j - \mu_j$, amendment costs or negotiated payments. For officers with that priority and a positive collection coefficient, the revenue-equivalent thresholds measure the additional public receipts needed to accept the less preferred outcome mix, separately for low-value and higher-value properties.

Scope of the empirical mapping. The preferred-bill weight, career recognition, amendment costs, expected collection shares and bargaining weights are not jointly estimated. Career beliefs describe views about advancement and reform support. Citizen choices concern offered hypothetical arrangements rather than observed negotiations. Record histories describe persistence and movement towards independent measurements. These sources assess components of the institutional comparison without recovering the complete payoff of an individual amendment.

5 Data

The analysis combines annual property-tax records from Lahore, Punjab Pakistan with independent remote-sensing measurements, physical site measurements and evidence from tax officials. These sources observe different parts of the administrative process. The annual records show what the department carries forward and changes. Remote sensing supplies a common physical reference for the same properties. Site visits allow the two measurement sources to be compared with an external reference. The officer datasets describe choices over administrative problems, preferred tax burdens, career beliefs and reform support. While the citizen survey's conjoint experiment describes choices over hypothetical scenarios of officers offering personal payments to reduce the bill, assessment errors and citizen preferences for accepting a particular set of profiles with a perosnal payment, assessment errors and final bill. The property identification number (PIN) connects property records across sources; officers are connected through their assessment circles.

5.1 Study site and geographic coverage

The property-level analysis covers sampled properties in Lahore's provincial property-tax register. Lahore District had 13.0 million residents in the 2023 census, compared with 11.1 million in 2017, with reported average annual population growth of 2.65% (Pakistan Bureau of Statistics, 2023). These district totals describe the broader setting; the study's geographic frame follows the provincial tax jurisdiction. Lahore Cantonment is outside that jurisdiction and is excluded from the property sample.

Within this setting, a common provincial assessment framework operates across neighbourhoods with different property values and physical layouts. Government surveys of housing show that the study site had approximately 1.74 million regular housing units in 2017 which grew to 2.01 million units by 2023. This scale of residential occupation and ongoing construction creates a continuing need to update property records. Physical layout also shapes what independent imagery can measure: parcel and building boundaries are easier to distinguish where structures are separated than in dense, contiguous development. Section 5.3 describes the resulting remote-sensing coverage and its observable differences from the wider register and field-validation sample.

The administrative data panel spans over the fiscal years 2022–23 to 2026–27, covering the January 2025 introduction of capital-value taxation; the independent imagery supplies a May 2026 reference. The following subsections describe the officer instruments and the separate field measurements. Official formula values, predicted market values and circle medians serve different purposes: the first reconstruct liability under administrative valuation rates, the second describe estimated property values, and the third describe the local property-value environment. A value band therefore refers to its stated measure and unit, rather than a household-income category.

5.2 Administrative property records

The Excise and Taxation Department, Government of Punjab, Pakistan, provided an anonymized cadastre of 1 million properties that includes information on property use (residential or commercial), ownership status (owned or rented), and property location (main or off-road). For each property-year, the cadastre further reports the name of the tax circle, neighborhood information stored as "locality", land area, covered area, number of storeys, and gross annual rental value. In addition, the cadastre contains information on the property's valuation category, which captures the quality of facilities and infrastructure in the property's locality. Each property is assigned a valuation category ranging from A to G. For fiscal years 2024 onwards, the cadastre now also reports applicable neighborhood-level valuation rates, the formula-implied property value, and an indicator identifying whether the property was billed as per the new capital-value taxation. Similarly, the data also contains information on total liability payable, any discounts applied and total amount recovered from each property. Of these 1 million properties, 581,386 are residential properties, which is the sampling frame for this paper. From this sampling frame, a sample of 22,054 properties was drawn from all 162 tax circles of Lahore. From these 162 tax circles, properties in these value bins were stratified by land area and were randomly selected above and below the median strata such that each circle had 130-140 properties.

An anonymized alphanumeric property-identification number (PIN) links the property panel across five years and independent measurements. Comparing the same property's entries reveals whether its recorded land or covered area changes, while comparing an entry with an independent reference reveals the direction and size of disagreement.

5.3 Independent remote-sensing measurements

The remote-sensing layer was assembled for 10,564 properties using the ESRI satellite imagery layer for Lahore from May 2026. For each selected property, an independent team of researchers delineated its parcel boundary. Land area was calculated from the parcel polygon. The layer

also contains the building footprint and height, and a predicted covered-area measure produced using a method described in Appendix D. The physical inputs can therefore be valued under the same official rate schedule as the administrative entries.

The remote-sensing observations link to the administrative records by PIN. A discrepancy compares two measurements of the same linked property, rather than relying on similarity in addresses or plot size. Because the imagery is observed in 2026, it provides a fixed cross-sectional reference. Changes in distance from that reference describe changes in recorded quantities; they do not require an annual satellite estimate of physical change.

The coverage reflects what the imagery can reliably distinguish. The original frame excludes Lahore Cantonment, which falls outside the department's jurisdiction, and high-density, contiguous and irregular building fabric where individual parcel and building boundaries are difficult to delineate. Its density screen uses WorldPop 2020 population cells and a threshold of 150 persons per 100-metre cell. These restrictions favour planned, organised and gated neighbourhoods with separable structures, a familiar issue in urban remote sensing (Kuffer et al., 2016).

The sample's observable composition makes this scope clear. Organised properties account for about 77% of the remote-sensing sample compared with 30% of the wider register. Selected properties have larger land and covered areas and higher tax liabilities than the non-selected register. Table F.1 retains the original comparison with both the non-selected register and its organised or gated segment. Normalised differences are substantially smaller within the organised segment. The organised segment is therefore the relevant coverage benchmark for the remote-sensing sample.

Table 5.1: Balance: RS sample against the organised/gated segment and the non-selected register

	RS	Organised/	Non-	Normalised difference	
	selected	gated	selected	vs org./gated	vs non-selected
Land area (sq. yd)	392.7	276.3	179.4	0.14	0.25
Covered area (sq. ft)	3,497	2,723	1,736	0.09	0.20
Tax liability (PKR)	9,670	7,986	4,184	0.09	0.32
Population density (/100 m)	118.6	117.3	114.8	0.08	0.21
Observations	10,564	156,983	546,108		

Group means. "Organised/gated" is the set of *non-selected* properties recorded as organised, planned, or gated, the segment the RS frame was drawn to cover; "non-selected" is the entire register outside the RS sample and includes peripheral low-density areas as well. Balance is read from the normalized difference, $(\bar{x}_{RS} - \bar{x}_g)/\sqrt{(s^2_{RS} + s^2_g)/2}$ (Imbens and Rubin, 2015), with |n.d.| > 0.25 signalling substantial imbalance (Imbens and Wooldridge, 2009; Imbens and Rubin, 2015). At this sample size every pairwise t-test rejects at $p < 0.001$, so no p-values are reported. Density is WorldPop 2020 per 100 m cell.

5.4 Field measurements

Trained enumerators visited properties and independently measured land and covered area. These observations are linked by PIN to the administrative records and the remote-sensing layer, allowing the same physical input to be compared across three sources. Field observations provide the external reference for comparing the two measurement sources.

The validation comparison contains 2,768 land observations and 2,504 covered-area observations with the required measurements. The analysis tracks recorded areas over time for

properties linked to the field survey that have positive area entries in each year examined. This includes 2,013 properties for land area and 2,009 for covered area. Recorded land area remained unchanged for 1,927 properties, while recorded covered area remained unchanged for 1,864 properties. Unchanged records do not necessarily agree with the field measurements. Each result reports the number of properties included in that comparison.

5.5 Estimated Market Values

I use estimated market values to compare properties on a common measure of economic value and examine how officers' preferred tax burdens and citizens' acceptance of paid concessions vary across the property-value distribution. These values are predicted using a random forest model trained on real estate agents' valuations of 12,363 properties, using property characteristics and location. They also allow me to describe the value composition of assessment circles. Estimated market values are distinct from official formula values, which apply prescribed valuation rates to recorded property attributes to determine tax liability. The market-value estimates provide a measure of property wealth, rather than household income, and are subject to prediction error. Appendix E details the method used for estimating market values.

Table 5.2 summarises the samples used in estimation, annual record linkage and coverage of the independent measurements. Each analysis uses the observations required for its comparison.

5.6 Survey Data

5.6.1 Frontline Tax Officers

Officers preferences and beliefs survey

The first wave of officers' survey was conducted in April - May 2025 and elicits preferred tax structures by presenting respondents with six residential properties, following an approach similar to Fisman et al. (2020). Figure F.1 shows the Android dashboard used for this exercise. It displays each property's lot size, built area, use, predicted market value, and number of stories. The displayed properties are randomly selected from three groups: low-value properties below the 50th percentile of the value distribution, medium-value properties between the 50th and 90th percentiles, and high-value properties above the 90th percentile. For each property, officers report what they believe its current average tax rate is and what they think it should be. Piloting identified Lahore's average tax rate as a useful benchmark for these questions. Officers first indicate whether the perceived current rate and their preferred rate are above or below this benchmark, then specify by how much. The progressivity analysis uses the final preferred annual tax liabilities to describe how each officer's preferred tax bill changes with property value.

The survey also asks which type of official makes better decisions and which is likely to have a successful career. For each question, officers choose among an official who follows personal interests, implements departmental direction, or listens to citizens. Asking these questions separately distinguishes beliefs about career advancement from views about how officials should make decisions. The survey also records officers' grade, education, time in their current circle, and expected promotion year.

Table 5.2: Estimation samples and annual record linkage

Sample	Properties	Percent of panel
A. Cadastre and retained administrative panel		
Lahore cadastre, all property uses	≈ 1,000,000	–
Residential cadastre	556,240	–
Retained administrative panel	22,054	100.00
B. Administrative records, FY2022–23 to FY2026–27		
FY2022–23 matched	22,054	100.00
FY2023–24 matched	22,054	100.00
FY2024–25 matched	22,054	100.00
FY2025–26 matched	22,054	100.00
FY2026–27 matched	22,054	100.00
Matched in all five years	22,054	100.00
Positive land entries in all five years	21,816	98.92
C. Independent measurements		
Full remote-sensing sample	10,564	–
Remote-sensing land linked to panel	9,206	41.74
Land: administrative, RS and field	2,797	12.68
Covered area: administrative, RS and field	2,703	12.26
Field-linked land histories	2,013	9.13
Field-linked covered-area histories	2,009	9.11
D. Estimation samples		
Land-correction eligible	9,030	40.94
Linked to officer priorities	7,964	36.11
Complete land-correction regression sample	7,143	32.39
With preferred-bill alignment	2,895	13.13
Collection regression sample	15,850	71.87
Fiscal and distributional sample	7,956	36.08
Fiscal comparison with both field measurements	2,407	10.91

Notes: Counts are distinct properties in the retained analysis files. Percentages use the 22,054-property administrative panel. Annual match rates describe linkage within this retained panel; the positive-land requirement is separate. Measurement-validation rows use the samples underlying Table 9.1. The full remote-sensing sample is reported before linkage. The balance comparison in Table F.1 uses its retained 10,132-record RS extract. Estimation samples overlap and are not additive.

Qualitative Surveys

The qualitative survey was conducted between May-June 2026 and asks frontline officers their information-gathering processes leading up to the capital value reform and how the reform changed their duties and functions. The interviews document the assessment and approval process, the location of discretion within the administrative hierarchy, and whether enforcement practices changed alongside the reform. The interviews describe field verification, documentation, supervisory approval, collection pressure and the continuation of old entries. They provide institutional evidence on the work required to amend the base.

Conjoint Experiment

The conjoint experiment, also conducted with qualitative surveys, presents tax officials with pairs of scenarios containing five randomly varying attributes and asks them to select the scenario. The attributes are i) compliance by low-value property owners; ii) compliance by medium- and high-value property owners; iii) overall support; iv) the fraction of properties that are delinquent; and v) the fraction of properties that are underassessed. To maximize statistical power to estimate the weights on these attributes, randomization is performed using a Bayesian Adaptive Choice Experiment (BACE) (Drake et al., 2024). Figure F.3 shows the exhibit from the experiment.

5.6.2 Taxpayers

Conjoint Experiment

I also conducted a separate face-to-face household-level survey with 2,100 taxpayers in Lahore (see Appendix F.3 for how citizens were selected). The survey included questions about taxpayer perceptions of tax administration and ease of getting an incorrect record corrected. Additionally, the survey included a conjoint experiment in which respondents choose between pairs of hypothetical property-tax assessment options. The exercise uses a Bayesian Adaptive Choice Experiment (BACE), with ten intended choice tasks per respondent. The bargaining-share calculation uses selected options that combine a human assessor, a positive payment to the assessor, and a positive reduction in the displayed tax bill (see Figure F.4).

For each selected deal, the officer's share is calculated by dividing the assessor payment by the tax reduction. The citizen's share is the remaining saving after the payment, divided by the same tax reduction. For example, a tax reduction of PKR 5,800 and an assessor payment of PKR 3,500 give the officer 60.34% of the saving and leave the citizen with 39.66%. Shares are first averaged across each respondent's selected paid deals and then across respondents within each administrative circle, giving each contributing respondent equal weight. Respondents who select no paid deals are excluded from this calculation.

These shares describe how the monetary saving is divided in respondents' chosen hypothetical deals. Under the Nash bargaining model, they also represent bargaining weights when both parties value monetary gains linearly, disagreement leaves the original bill payable and the officer unpaid, and there are no additional costs or benefits. The exercise therefore provides a conditional measure of surplus sharing, rather than evidence of payments made in actual tax administration.

6 Inherited Distortions and Record Persistence

The main empirical results concern the inherited register: earlier recording incentives left a detectable imprint, and many unchanged entries still disagree with current field measurements. The historical exemption covered one house used for residential purposes with a land area of no more than 1,125 square feet. The exemption rule applied to all neighborhood types except Category A localities—the highest rental-value category under the old property-tax system. The historical exemption comparison examines how earlier incentives shaped recorded placement. The field and panel comparisons then establish whether entries carried forward still disagree with independent measurements. Together they distinguish the origin of favourable recording

from the subsequent task of maintaining accurate inputs.

6.1 The imprint of the historical exemption

I first show descriptively that, under the previous tax regimes, officers exploited the exemption rule. In Figure 6.1, 58.7% of the weighted non-A sample is recorded between three and five marla, compared with 12.7% in Category A. All these properties are independently measured above five and up to eight marla. In other words, properties measured above the old exemption cutoff are much more often recorded within the exemption range in the localities where the exemption applied.

Panel A shows this comparison. The horizontal axis is recorded land area; the height of each bar is the weighted share in that area interval. Red bars identify entries between three and five. Properties independently measured closer to five receive more weight. Panel B reports non-A minus Category-A differences of 33.7, 45.5 and 45.9 percentage points for properties independently measured above five and up to six, seven and eight marla, respectively. The positive differences mean that recording within three to five marla is more common in non-A localities across all three ranges. These comparisons include only properties independently measured above five.

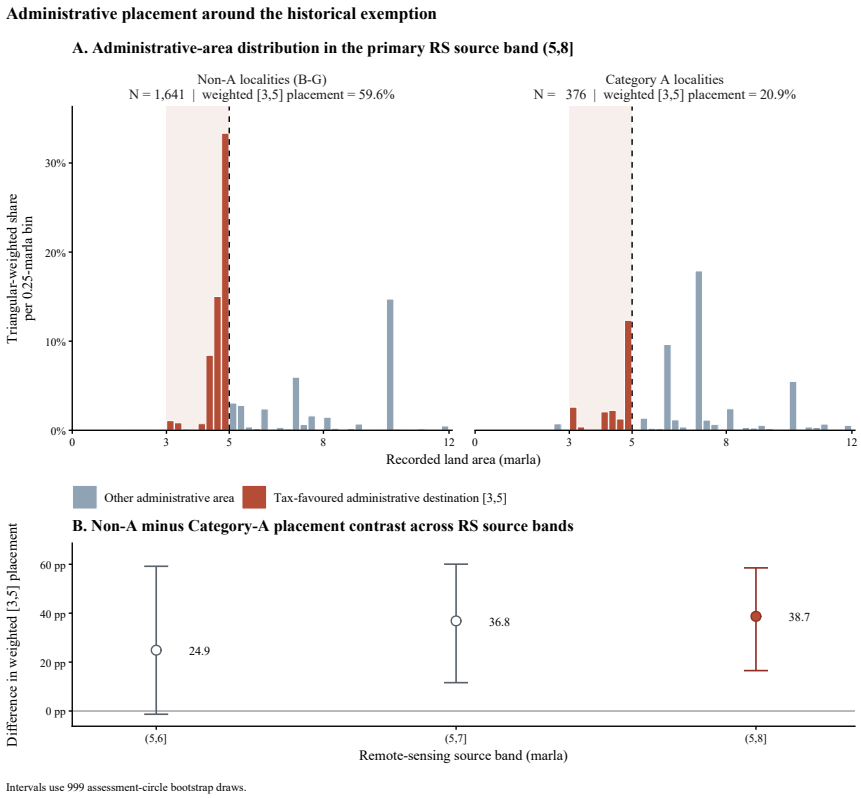


Figure 6.1: Administrative placement around the historical exemption. Panel A shows placement for RS land area in $(5, 8]$. Panel B shows non-A minus Category-A placement in $[3, 5]$ for source bands $(5, 6]$, $(5, 7]$ and $(5, 8]$, with $N = 717$, 1,315 and 1,893. Vertical bars are 95% assessment-circle bootstrap intervals. These upper-side comparisons differ from the at-or-below-five outcome in Table 6.1.

I then estimate the difference between recording discontinuities around the exemption

threshold. Let L_i denote recorded land area from administrative data and $x_i = L_i^{\mathrm{RS}} - 5$ RS land area centred at five marla. Define $Y_i^5 = \mathbf{1}\{L_i \le 5\}$ and $I_i^5 = \mathbf{1}\{x_i > 0\}$. Locality group g is Category A or non-A (Categories B–G). For each group, the right-minus-left discontinuity is

$$\delta_g = \lim_{x \downarrow 0} \mathbb{E}[Y_i^5 \mid x_i = x, g] - \lim_{x \uparrow 0} \mathbb{E}[Y_i^5 \mid x_i = x, g], \qquad d_5 = \delta_{\mathrm{nonA}} - \delta_{\mathrm{A}}.$$

The separate local-linear fits have the form

$$Y_i^5 = \alpha_g + \delta_g I_i^5 + b_{gL} x_i (1 - I_i^5) + b_{gR} x_i I_i^5 + \Gamma_g'(Z_i - \bar{Z}_w) + \varepsilon_i. \tag{6.1}$$

Here i indexes properties and g denotes Category A or non-A. The binary outcome Y_i^5 equals one when recorded land is at or below five marla; x_i is land area estimated using remote-sensing technology (RS) minus five, in marla, and I_i^5 equals one above that RS cutoff. The intercept α_g is the fitted level immediately below the cutoff at the common reference $\bar{Z}_w$, and δ_g is the right-minus-left jump in that level. The coefficients b_{gL} and b_{gR} are the slopes in RS area below and above the cutoff. The baseline Z_i contains only the main-road indicator, centred at its pooled triangular-weight mean within the estimation window, $\bar{Z}_w = 0.5045$. The coefficient Γ_g is group-specific, and ε_i is the unexplained property-level component. Centring changes the reference for the fitted levels, not the estimated jump. The difference $d_5 = \delta_{\mathrm{nonA}} - \delta_{\mathrm{A}}$ is the reported contrast between recording discontinuities.

The primary specification uses a common two-marla bandwidth and main-road status only. Uncertainty is calculated by drawing assessment circles with replacement 999 times, retaining all included properties in each selected circle and re-estimating both jumps together. The middle 95% of the resulting differences gives the percentile confidence interval. Drawing both groups together preserves their shared circle-level variation (Efron and Tibshirani, 1986; Cameron and Miller, 2015). These bootstrap intervals accompany the bias-corrected point estimates; they are distinct from the robust analytical intervals proposed by Calonico et al. (2014).

Table 6.1 compares the fitted recording probabilities immediately below and above five in the independent measurement. As independently measured land area through remote sensing crosses from just below to just above five marla, the estimated probability of being recorded at or below five falls by 22.33 percentage points in Category A. In non-A localities, where the old exemption applied, it instead rises by 33.72 percentage points. The difference between these two jumps is $d_5 = 33.72 - (-22.33) = 56.05$ percentage points (bootstrap s.e. 14.66; 95% percentile confidence interval [20.03, 79.06]). This shows that, as compared to type A-category localities, properties measured as greater than 5 marla using remote sensing are 56 percentage points more likely to appear below 5 marla in the administrative records for non-A localities.

Panel B of Table 6.1 asks a related question: are properties unusually likely to be recorded exactly at five marla, compared with nearby round numbers? For each comparison point—four, five and six marla—I calculate the share recorded exactly at that point among properties independently measured above it. I then compare the share at five with the average of the shares at four and six. In non-A localities, the share at five is 13.35 percentage points higher using remote-sensing measurements and 7.84 points higher using field measurements. The corresponding Category-A estimates are 3.44 and 5.17 percentage points. The RS Category-A interval [0.56, 9.57] percentage points excludes zero, whereas the field interval [0.00, 18.37] includes zero. Concentration at five is therefore more pronounced in non-A localities, but is not confined to them. Each comparison uses its own group of properties, with counts reported in the table. This concentration at five, the former exemption cutoff, provides further evidence consistent with the historical recording incentive.

Conventional estimation also gives a positive non-A jump (30.52 percentage points). Appendix G.1 reports the fitted cutoff levels and robustness diagnostics.

Table 6.1: Changes in recording around five marla and concentration exactly at five

Panel A. Recorded area at or below five marla					
Group	BC jump	Boot s.e.	95% interval	N_-	N_+
Category A	−0.223	0.126	−0.427 to 0.112	95	264
Non-A	0.337	0.070	0.197 to 0.472	1,889	1,105
Non-A minus A	0.560	0.147	0.200 to 0.791	1,984	1,369
Panel B. Extra share recorded exactly at five marla					
Benchmark / group	Estimate	Boot s.e.	95% interval	$N_4/N_5/N_6$	
RS / non-A	0.133	0.035	0.070 to 0.204	1451/1461/1719	
RS / A	0.034	0.024	0.006 to 0.096	302/345/290	
Field / non-A	0.078	0.028	0.033 to 0.140	2443/1272/1871	
Field / A	0.052	0.050	0.000 to 0.184	109/58/304	

Jump estimates are in probability units: 0.10 is 10 percentage points. N_- and N_+ count properties below and above the cutoff in independent area. Panel B subtracts the average recording share at four and six from the share at five. Outcome: MIS 2021–22 recorded land area at or below five marla. Right-minus-left jumps from separate locality-group local-linear fits with triangular weights, common bandwidth 2 marla, quadratic bias correction and a main-road indicator. Log RS covered area is excluded from the baseline. Category-A circle counts below/above are 13/23, and non-A counts are 56/43; circles can contribute on both sides. Counts show effective support; points exactly at five enter the left side, and the lower zero-weight boundary is excluded. Assignment cells screen mixed categories but are not fixed effects. Uncertainty uses 999 joint circle resamples (seed 20261018); components can differ in the final decimal because of rounding. Panel B uses 999-circle-bootstrap excess-landing estimates: the landing rate at five minus the mean rates at four and six. Its three denominators are the distinct source-band counts at each comparison point; they must not be summed as unique properties.

6.2 Persistence of unchanged records

Figure 6.2 shows how recorded land and covered areas change between FY2022–23 and FY2026–27 relative to a fixed remote-sensing (RS) benchmark. The gap is defined as RS log area minus administrative log area. Each series starts at zero because the figure plots changes in this gap relative to FY2022–23, rather than the gap itself. Since the RS measurement is held fixed, an upward movement means that recorded area has decreased, while a downward movement means that recorded area has increased.

The main finding is that recorded areas change little over the five years. The solid line shows a small increase in the land-area gap, with the largest movement occurring in FY2026–27. In the balanced sample of 9,140 properties across 77 assessment circles, the gap increases by 0.0108 log points between FY2022–23 and FY2026–27 (95% circle-clustered confidence interval: $[0.0048, 0.0169]$). This corresponds to an average decline of approximately 1.1% in recorded land area on the log scale. The mean gap moves from −0.060 to −0.049, becoming slightly less negative. Thus, the change is statistically distinguishable from zero but small, and the average recorded land area remains above the fixed RS benchmark.

The dashed line shows a similarly small movement in covered area. Its gap falls slightly through FY2025–26 before returning towards its starting level in FY2026–27. This pattern indicates a small initial increase in recorded covered area followed by a partial reversal. Both

series therefore show limited average updating of recorded physical quantities.

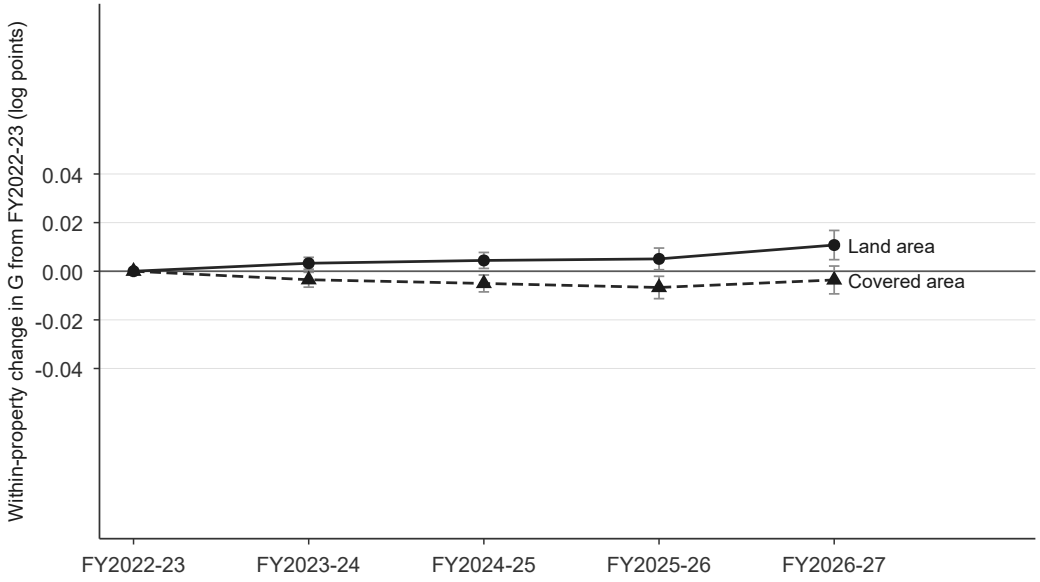


Figure 6.2: **Changes in recorded-area gaps relative to FY2022–23.** The figure plots average within-property changes in $G_{it,k} = \log X_{i,k}^{\mathrm{RS}} - \log X_{it,k}^{\mathrm{admin}}$, where k denotes land or covered area. Both series are normalised to zero in FY2022–23. RS measurements are fixed across years, so positive changes indicate decreases in recorded area and negative changes indicate increases. The solid line represents land area; the dashed line represents covered area. The figure describes changes in administrative records and does not identify a reform effect.

These small average changes are consistent with the continued use of inherited physical quantities under the new tax formula. Where an inherited quantity is inaccurate, applying a new formula can carry that discrepancy into current tax liability.

6.3 Unchanged records often differ from current field measurements

Table 6.2 asks whether records that remain unchanged over time deviate from current measurements conducted in field visits by trained enumerators. The comparison links the five-year administrative histories to the field measurements and reports the prevalence and concentration of substantial discrepancies.

The discrepancy ratios show whether unchanged records match current property dimensions. A record is classified as unchanged when its area entry is the same in every year examined. A discrepancy is classified as substantial when the administrative records deviate from field measurements by more than 25%.

Among unchanged records linked to the field survey, 45.0% of land entries have a substantial discrepancy. The corresponding share for covered area is 38.7%. Thus, almost half of the checked land records remain unchanged despite differing substantially from the current field measurement. The final row asks the question from the other direction: of all substantial discrepancies, how many occur in unchanged records? The answer is 94.5% for land and 90.4% for covered area. Most substantial discrepancies therefore occur in records carried forward without change. These findings show that many unchanged records do not accurately describe current property dimensions.

Table 6.2: Discrepancies in unchanged records

	Land area	Covered area
Field-linked histories	2,013	2,009
Unchanged histories	1,927	1,864
Discrepancy rate among unchanged histories	45.0%	38.7%
Share of all discrepancies in unchanged histories	94.5%	90.4%

Notes: The samples link field measurements to five positive annual administrative entries. An unchanged history has identical recorded area in all five years. A substantial discrepancy is an absolute administrative–field log difference greater than 0.25. The discrepancy rate divides discrepant unchanged histories by all unchanged histories; the final row divides them by all discrepant histories. Field measurements are an external reference.

7 Administrative Costs and the Priority Given to Recovery

I use the Bayesian adaptive choice experiment (BACE), an approach to eliciting preferences through choices adapted to previous responses (Drake et al., 2024), to examine the administrative preference described in Section 4.4. Officers select the outcome they judge most likely to continue functioning effectively over the next three to five years. The comparisons measure the relative valuation of less underassessment and less delinquency under this effectiveness question, and express the trade-off in revenue units. They inform the model's distinction between assessment and recovery outcomes.

7.1 Recovery priority at fixed collections

In BACE, officers repeatedly choose between two hypothetical administrative outcomes. Each alternative specifies collection percentages for low-value and higher-value properties, public support, underassessment and delinquency. Underassessment means that the assessed bill is too low; delinquency means that an assessed bill remains unpaid. At fixed collections and public support, the comparison identifies the relative preference for these two outcomes. Let β_{Uj} and β_{Dj} be the fitted coefficients on underassessment and delinquency. Holding collections and public support fixed, one percentage point more underassessment contributes β_{Uj} to the change in utility, while the accompanying one point less delinquency contributes $-\beta_{Dj}$.[2] Because the experiment changes these two attributes together, the choices identify how officers value this exchange. The combined change is their difference, $\beta_{Uj} - \beta_{Dj}$:

$$\omega_j = \beta_{Uj} - \beta_{Dj}, \qquad W_j = \mathbf{1}\{\omega_j > 0\}. \tag{7.1}$$

The indicator W_j equals one when $\omega_j > 0$ and zero otherwise. A value of one therefore implies $\beta_{Uj} > \beta_{Dj}$, showing that officers favor less delinquency over more underassessment, at the same collections and public support. I call this preference *recovery priority*. A value of zero includes both the opposite preference ($\omega_j < 0$) and indifference ($\omega_j = 0$). The null of equal valuation and the alternative of different valuations are

$$H_{B,0} : \omega_j = 0, \qquad H_{B,A} : \omega_j \neq 0. \tag{7.2}$$

[2]Underassessment U and delinquency D move one for one, with $U + D = 76\%$. 76% represents average compliance from the study site for the year 2024 calculated using admininstrative data provided on tax receipts.

Recovery priority does not imply that the officer positively values underassessment. The officer may dislike both problems but dislike delinquency more, making an increase in underassessment acceptable in return for less delinquency. The fitted choice probabilities describe selection under the effectiveness question using the standard logit framework (Train, 2009).

Holding collections and public support fixed, I calculate each officer's fitted probability of choosing ten percentage points less delinquency together with ten points more underassessment: $\pi_j = \text{logit}^{-1}(10 s_j \omega_j)$, where $s_j > 0$ is the fitted choice scale. A probability above one half favours less delinquency; a probability below one half favours less underassessment. Figure 7.1 shows these probabilities across officers. This measure combines the contrast with its choice scale instead of comparing raw utility magnitudes across officers.

Figure 7.1 shows that for 174 of 289 officers (60.2%), the estimated preference favours less delinquency even when it comes with more underassessment. The remaining 115 officers (39.8%) favour the reverse exchange. Thus, a majority favours reducing unpaid billed tax at a given level of collections.

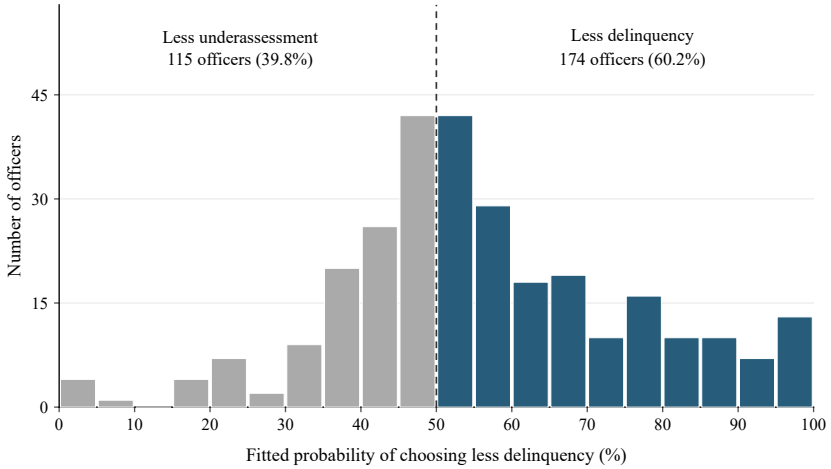


Figure 7.1: Recovery priority at equal collections. Each of the 289 officers contributes a fitted probability of choosing ten percentage points less delinquency together with ten points more underassessment, holding collections from both property groups and public support fixed. Bars count officers in five-percentage-point probability intervals. The dashed line marks 50%: values to its right favour less delinquency, while values to its left favour less underassessment. Probabilities use the saved coefficient means and describe choices under the experiment's administrative-effectiveness question.

The results connect to the model's comparison of correction and recovery at equal receipts in Section 4.3. When both tasks leave positive uncollected demand, equation (4.12) gives their fiscal payoff difference as $(\mu_j - a_j) g_i$, where $g_i > 0$ is the additional bill created by correction, μ_j rewards assessed liability and a_j penalises unpaid demand. The fiscal component favours recovery when $a_j > \mu_j$: the penalty attached to adding demand outweighs the credit for establishing the additional liability. The preference of 60.2% of officers for less delinquency at equal collections is consistent with this direction of preference. It shows why equal receipts need not make the two outcomes equally attractive.

The connection to record maintenance follows from the full correction condition. In equation (4.13), the penalty on the uncollected part of an additional bill reduces F_{ij}, the officer's gain from correction rather than retention before any private payment. Under the model's bargaining

assumptions, a feasible upward correction proceeds when $F_{ij} \geq g_i$, as stated in equation (4.11). BACE provides evidence that greater concern about unpaid demand can make this condition harder to meet when the additional bill is not fully collected. Section 7.2 measures how much additional revenue would offset the less preferred exchange in the fitted choices.

7.2 Compensating public revenue

For an officer with recovery priority, less underassessment accompanied by more delinquency is unattractive when collections are unchanged. I calculate how much extra public revenue would offset this trade-off for one percentage point less underassessment and one point more delinquency, holding public support fixed. At that amount the officer is indifferent; additional revenue above it makes the outcome with less underassessment more attractive. The amount is extra revenue for the government, not a payment to the officer.

I obtain the revenue equivalents from the fitted BACE utility specification

$$v_j = \beta_{Lj} C_L + \beta_{Hj} C_H + \beta_{Sj} S^{pub} + \omega_j U. \tag{7.3}$$

Here C_L and C_H are collection percentages for low-value and higher-value properties, S^{pub} is public support, and U is underassessment. The β coefficients weight the collection and support attributes, while ω_j is officer j's relative valuation of underassessment and delinquency, defined above. Appendix **??** explains identification and derives the revenue equivalents from differences between options.

Table 7.1 reports that officers need to raise an additional median compensating amount of PKR 4.96 million from low-value properties and PKR 13.85 million from higher-value properties to prefer correction over recovery. While preferences vary by officers, results show that recovery priority can only be outweighed by sufficiently large and feasible revenue gains from correcting records in the fitted choices.

Table 7.1: Median revenue equivalents for less underassessment

Sample / collection group	N	$\omega > 0$ (%)	Median	Middle half of estimates
Group-specific sample / Low	212	64.2	4.96	−4.97 to 16.41
Group-specific sample / High	221	62.0	13.85	−19.62 to 45.63
Same officers / Low	178	64.6	4.85	−4.82 to 17.61
Same officers / High	178	64.6	15.96	−14.18 to 45.63

Notes: Amounts are PKR millions for one percentage point less underassessment and one point more delinquency. The full sample has 289 officers. Group-specific samples require a positive estimated collection coefficient, excluding 77 (26.6%) and 68 (23.5%) officers. The common sample requires both coefficients to be positive, excluding 111 (38.4%). Entries are individual ratios evaluated at posterior means. Within these samples, $\hat{\omega}_j > 0$ is equivalent to a positive revenue equivalent. The middle half describes variation across fitted officer preferences. The estimates concern outcomes judged likely to function effectively over three to five years. Appendix G.3 reports denominator screens and uncertainty.

8 Preferred Tax Burdens and Private Returns

The evidence relates to three components of the model: the official bill an officer favours, the tax saving available from retaining a lower assessment, and the terms on which households accept

a paid concession. The officer exercise describes preferred bills P_{ij}. The record comparison provides an RS-based benchmark for positive retention savings g_i. The citizen experiment describes acceptance and the division of savings in hypothetical concessions.

8.1 The schedule officers prefer

Officers' preferred average tax rates rise with predicted market value, suggesting a progressive allocation of the official burden. Let M_i denote the predicted market value of a property shown to an officer, and let P_{ij} denote the annual tax bill officer j thinks that property should pay. The preferred average tax rate, expressed as a percentage, is

$$\mathrm{ATR}_{ij}^{P} = 100\frac{P_{ij}}{M_i}. \tag{8.1}$$

Figure 8.1 summarises the pooled officer–property responses using LOWESS against log predicted market value, with bandwidth 0.7. The rising curve indicates that higher-value properties are assigned larger preferred tax bills relative to their value. It describes the pooled pattern in reported rates, rather than a common schedule imposed on every officer.

In the model, these preferred bills enter the correction payoff through $q_{ij} = |T_i^0 - P_{ij}| - |T_i^1 - P_{ij}|$. Correction is more attractive when it brings the assessed bill closer to P_{ij}. Progressive preferences therefore support correction where understated bills fall below the amounts officers consider appropriate and correction reduces that distance. The effect depends on the positions of the retained and corrected bills relative to the preferred amount, rather than on progressivity alone.

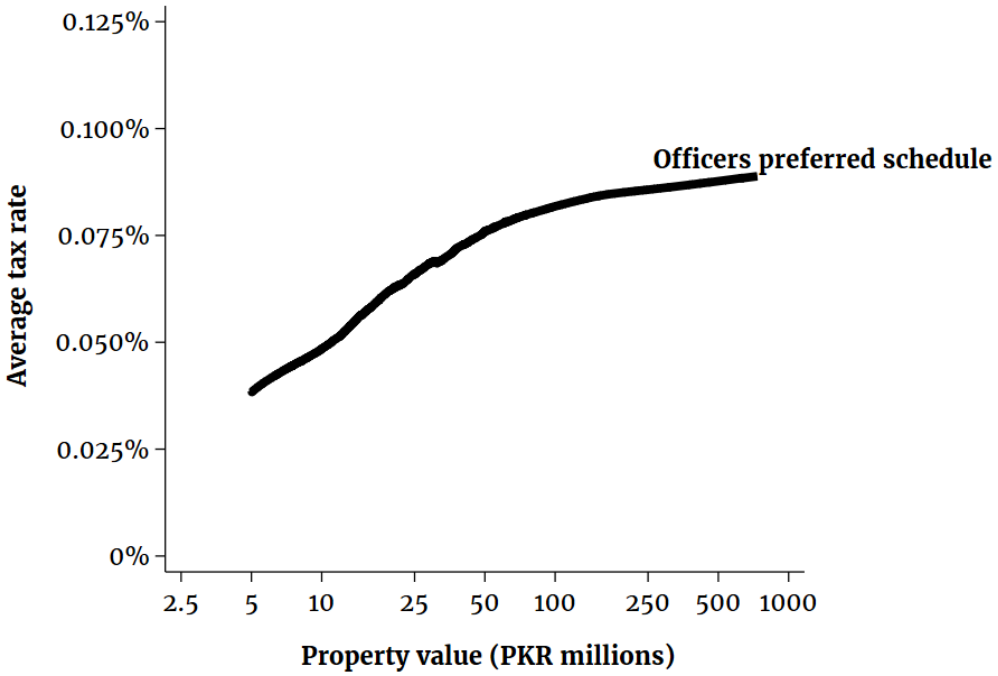


Figure 8.1: Officers' preferred average tax rates by predicted market value

Notes: The curve is a LOWESS smoother of preferred average tax rates against log predicted market value, with bandwidth 0.7. Preferred rates are annual preferred tax liabilities divided by predicted market value, expressed as percentages. The horizontal axis displays predicted market value in PKR millions on a logarithmic scale. The curve summarises pooled officer–property responses.

8.2 Where potential tax savings are concentrated

Properties above PKR 25 million in RS formula value account for 48.0% of total potential tax savings while representing 11.2% of the weighted property population. The comparison uses the existing 7,956 properties in 65 assessment circles and their inherited transport weights. It holds the official rates and capital-value schedule fixed and replaces recorded land and covered area jointly with RS inputs. For property i, define the potential annual saving from retaining the lower administrative-input assessment as

$$S_i = \max\{T_i^{\mathrm{RS}} - T_i^{\mathrm{Admin}}, 0\}.$$

Here T_i^{RS} and T_i^{Admin} apply the same formula to the alternative physical inputs. A property contributes a positive saving only when the RS-input bill exceeds the administrative-input bill. Lower RS-input bills contribute zero to this measure; their reductions are retained separately in the fiscal accounting. This prevents downward revisions elsewhere from cancelling the positive stakes in lower recorded assessments. The measure is a formula-implied liability difference, not a comparison of observed payments or receipts.

Figure 8.2 compares each formula-value group's share of properties with its share of total potential savings. The PKR 5–10 million group accounts for 48.5% of properties and 28.1% of savings; the PKR 10–25 million group accounts for 29.2% and 23.9%, respectively. The above-PKR-25-million group therefore accounts for a much larger share of savings than of properties. The absence of positive savings in the lowest group follows from the formula's exemption through PKR 5 million; it does not establish that those records contain no physical understatement.

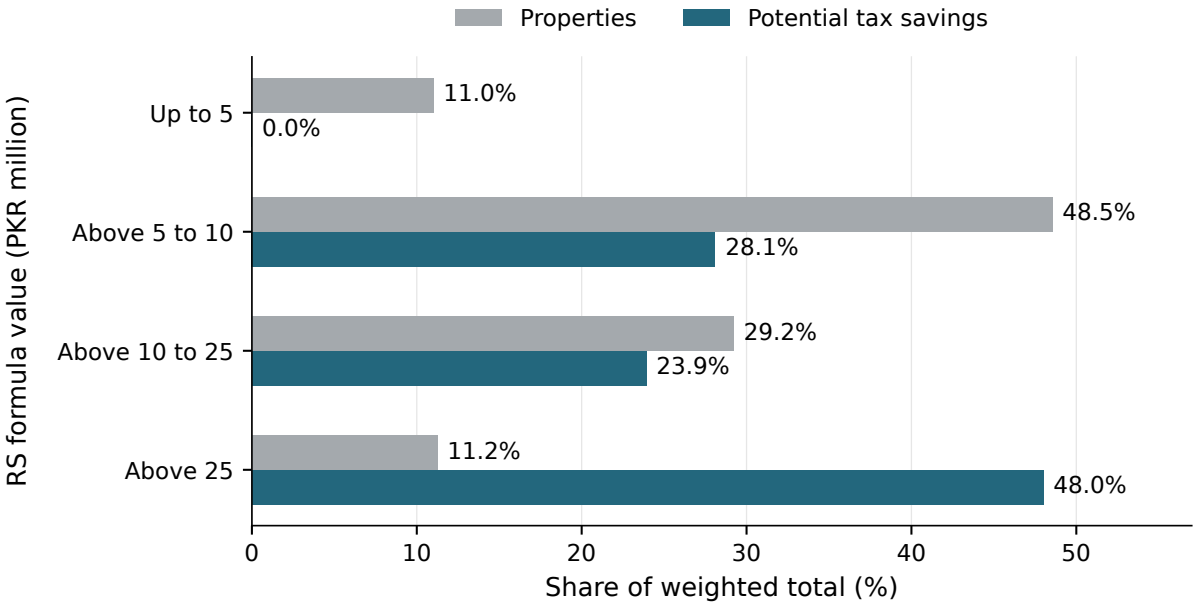


Figure 8.2: Concentration of potential tax savings from lower recorded assessments

Notes: The sample contains 7,956 properties in 65 assessment circles. Potential annual tax saving is $S_i = \max\{T_i^{\mathrm{RS}} - T_i^{\mathrm{Admin}}, 0\}$, where both liabilities apply the same capital-value formula to their respective land and covered-area inputs. For group b, the savings share is $\sum_{i \in b} w_i S_i / \sum_i w_i S_i$ and the property share is $\sum_{i \in b} w_i / \sum_i w_i$, using the inherited transport weights w_i. Groups are defined by official formula values calculated from RS inputs, rather than predicted market values. Bars report descriptive shares; confidence intervals for these shares have not been estimated. The comparison includes predicted covered area and does not establish verified evasion.

These amounts describe potential savings relative to the RS benchmark. The ranking and

the RS bill also use the same area inputs, so shared measurement error can affect the apparent concentration. The results locate monetary stakes under the stated comparison; they do not identify intentional understatement or verified evasion.

8.3 How selected concessions divide tax savings

The citizen exercise measures the division of a saving conditional on selecting a paid concession. For a selected human-assessor option with a positive payment and tax reduction, the assessor's share is the payment divided by the reduction in official liability. A PKR 5,800 reduction accompanied by a PKR 3,500 payment allocates 60.34% to the assessor and leaves 39.66% with the taxpayer. Appendix B provides the maintained surplus-sharing interpretation.

The figure contains 1,991 contributing respondents in 51 circles. Shares are averaged first across selected paid concessions within respondent and then equally across contributing respondents within circle. Each point in Figure 8.3 is one circle, positioned at the median predicted market value of its linked RS properties. The vertical intervals retain the original respondent-bootstrap uncertainty. Circle medians use 7,916 properties in the available RS valuation sample and are not restricted to respondents selecting concessions.

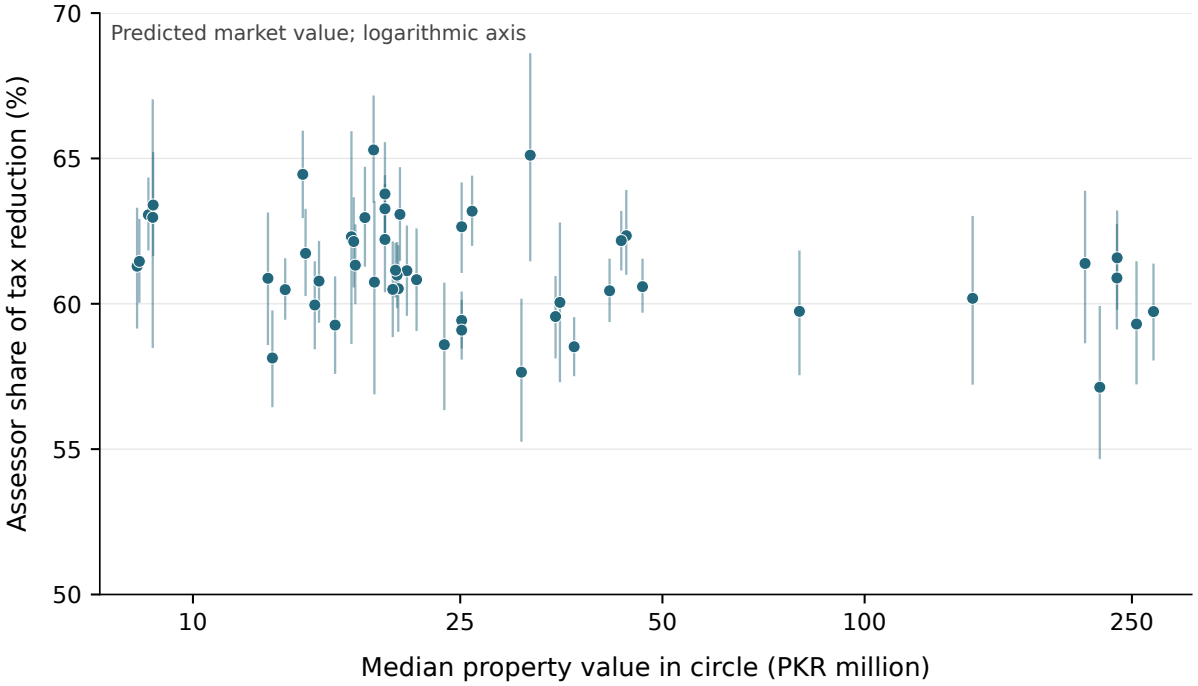


Figure 8.3: Assessor shares in selected hypothetical paid concessions

Notes: Each point represents one assessment circle. The assessor share is payment to the assessor divided by the reduction in official tax liability, averaged first across selected paid concessions within respondent and then equally across contributing respondents within circle. The figure contains 1,991 respondents in 51 circles. Vertical bars are the saved 95% percentile intervals from 4,999 respondent resamples within each circle, retaining complete choice histories. The horizontal axis reports circle median predicted market value in PKR millions on a logarithmic scale; 7,916 linked RS properties contribute to these medians. The intervals condition on the medians and exclude valuation uncertainty. These are selected hypothetical concessions, not observed payments.

The points show variation in the assessor's share across circles, with overlapping confidence intervals. A share of 60% means that PKR 60 of a selected PKR 100 tax reduction goes to the assessor and PKR 40 remains with the taxpayer. The figure describes a conditional division of

savings. It does not show where total payments are concentrated or how often such arrangements occur in actual administration. Its horizontal axis is predicted market value, whereas Figure 8.2 groups properties by RS official formula value; the two value measures and units of observation must remain distinct.

8.4 Acceptance of paid concessions across property values

Households with higher predicted property values accept hypothetical paid concessions more often. Figure 8.4 relates each respondent's acceptance rate to the predicted market value of their own property. The sample contains 1,919 respondents and 12,267 choices offering a paid inspector concession alongside an unpaid alternative. For each respondent, the acceptance rate is the share of these choices in which the paid concession is selected. Respondents receive equal weight, including those who reject every eligible concession. The mean acceptance rate is 51.3% in the lowest property-value quarter and 68.3% in the highest, a difference of 17.1 percentage points. The two middle quarters each have an acceptance rate of approximately 59.2%.

The grey dots show individual respondents' acceptance rates. The black line connects mean acceptance rates across ten property-value groups, plotted at each group's median value. Acceptance is generally higher towards the upper end of the value distribution, although it does not rise at every step. This comparison adds the frequency of acceptance to Figure 8.3, which describes the assessor's share conditional on a paid concession being selected.

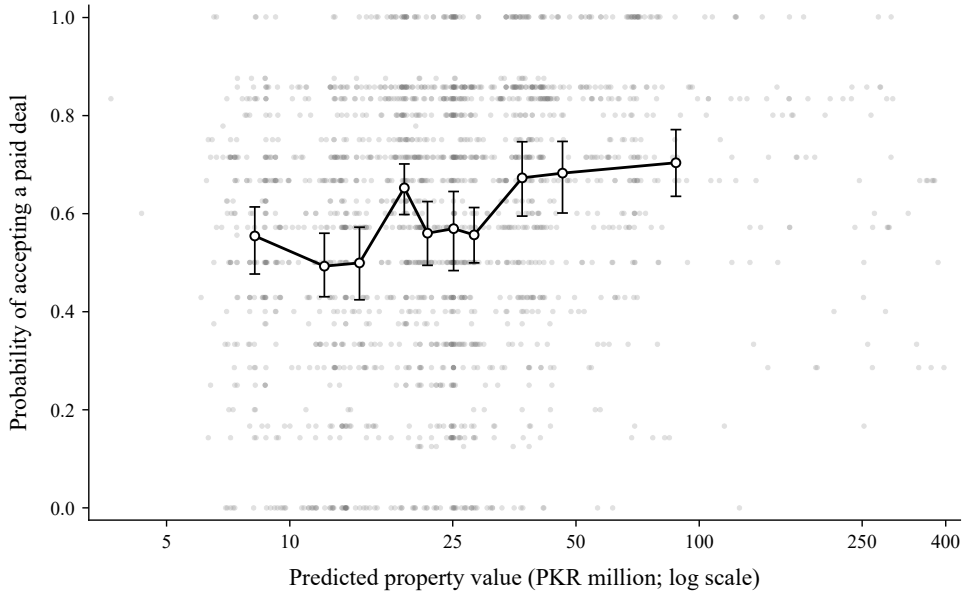


Figure 8.4: Acceptance of hypothetical paid concessions by property value

Notes: Grey dots show respondent-level acceptance rates; open circles show equally weighted means within predicted-market-value deciles, positioned at the median value in each group. The line connects these descriptive means. Values refer to respondents' own linked properties, in PKR millions on a logarithmic axis; tied values remain together. The sample contains 1,919 respondents and 12,267 completed choices with one paid inspector concession and one unpaid alternative, which may involve an inspector or automated assessment. A paid concession has a positive informal payment and a positive displayed tax reduction. Each respondent's rate is the share of eligible choices in which that concession is selected; 97 never-acceptors remain included. Tasks offering two paid concessions or no paid concession are excluded. Whiskers are 95% pointwise percentile intervals from 4,999 bootstrap draws, resampling 57 known circles and treating 35 respondents without circle linkage as individual sampling units. Complete respondent histories are retained; property values, saved offers and group boundaries remain fixed. Offers are adaptive, so the association may reflect both preferences and differences in the offers shown. The choices are hypothetical, not observed payments.

8.5 Implications for the private-return mechanism

The results connect preferred official burdens, potential savings and household participation to distinct components of the model. The preferred-rate pattern describes the bills P_{ij} officers favour. Where correcting an understated record moves the bill closer to that preferred amount, $q_{ij} > 0$ raises the officer's correction payoff F_{ij}. Progressive preferences can therefore strengthen the incentive to correct higher-value properties when the corrected bills better match the officer's preferred allocation.

That incentive can coexist with a paid agreement to retain the lower bill. In the model, $b_{ij}^{x*} = F_{ij} + \beta_{x,j}^{B}(g_i - F_{ij})$ when $0 < F_{ij} < g_i$. A higher correction payoff raises the payment required to compensate the officer, while a larger tax saving gives the taxpayer more to offer. If $F_{ij} \geq g_i$, the saving cannot support a mutually beneficial payment and correction proceeds. The concentration result locates substantial potential savings among higher-value properties: those above PKR 25 million account for 40.1% of savings and 18.0% of weighted properties.

The citizen results describe the other side of this potential agreement. Higher-value households accept the offered hypothetical paid concessions more often, and the selected concessions allocate part of the tax saving to the assessor. Acceptance, conditional payment shares and the amount saved describe different margins; a similar percentage share can imply a larger payment when the underlying saving is larger. These comparisons do not jointly measure realised officer

receipts.

The model thus accommodates progressive preferred burdens alongside paid retention of lower assessments. Private payments can compensate an officer for foregoing a correction that would otherwise be preferred. This interpretation links the evidence to the model's correction and participation conditions; it does not require officers to choose progressive preferences in order to obtain private returns. The model takes preferred bills as given.

9 Technology and the Verifiability of the Tax Base

The preceding results identify private returns and administrative costs that can favour retaining existing records. Independent measurement can strengthen the supervisors' oversight of delegated assessment by making recorded property characteristics easier to verify. Technology changes the information available to supervise this work. This section evaluates the accuracy of independent measurements, examines their usefulness for targeting verification, and assesses how replacing recorded inputs changes the distribution of liabilities under the same statutory schedule.

9.1 Relative Measurement Performance

Independent measurement provides a useful check on delegated assessment when it improves information about the tax base. I compare administrative and remote-sensing measurements with the same field reference, separately for land and covered area. For source m, mean squared error is $\mathrm{MSE}(m) = N^{-1} \sum_i (X_i^m - X_i^{field})^2$. It gives larger discrepancies greater weight. Relative improvement is $\lambda = 1 - \mathrm{MSE}(RS)/\mathrm{MSE}(Admin)$, so a positive value indicates lower squared error under remote sensing.

Table 9.1: Measurement error by source

	Attribute	
	Land area	Covered area
Properties in the validation sample	2,797	2,703
Tax circles	60	58
Mean squared error, administrative record	734.9	17,707,489.1
Mean squared error, remote sensing	317.8	9,448,461.8
δ_{RS} = MSE(RS) − MSE(admin)	-417.0***	-8,259,027.3***
	(151.7)	(2,530,600.6)
Median absolute error, administrative record	1.72	642.50
Median absolute error, remote sensing	0.41	482.76
λ = 1 − MSE(RS)/MSE(admin)	0.568	0.466
95% CI (circle cluster bootstrap)	[0.344, 0.741]	[0.268, 0.703]
λ in logs (robustness)	0.284	-0.038

The benchmark is the field measurement collected by trained enumerators on site. Accuracy is stated relative to that protocol; neither the site measurement nor remote sensing is assumed to equal latent physical truth without error.

δ_{RS} is the coefficient on a remote-sensing indicator in a paired squared-loss regression with property fixed effects, $L_{im} = \alpha_i + \delta_{RS}\mathbf{1}\{m = RS\} + u_{im}$, estimated on two observations per property. The property fixed effect makes δ_{RS} the mean within-property difference in squared loss, which is identically MSE(RS) − MSE(admin). The omitted source is the administrative record. A negative δ_{RS} means remote sensing is the more accurate source. Standard errors in parentheses, clustered by tax circle.

The interval for λ uses a percentile bootstrap resampling tax circles with replacement, 2,000 replications, seed 20260825. The log specification reweights away from the largest properties and is undefined at zero, so it drops 1 land and 13 covered-area properties.

Zero administrative or RS values are retained in the levels analysis. Land area is in marla and covered area in square feet, so the squared errors are not comparable in level across the two columns; λ is scale-free and is comparable.

For land area, the validation sample contains 2,797 properties across 60 assessment circles. MSE falls from 734.9 in the administrative record to 317.8 under remote sensing. The relative improvement is $\widehat{\lambda}_{land} = 0.568$, with a 95% assessment-circle bootstrap interval of $[0.344, 0.741]$. Remote sensing therefore reduces land-area MSE by approximately 57%. Median absolute error also falls, from 1.72 to 0.41 marla.

For covered area, the sample contains 2,703 properties across 58 assessment circles. MSE falls from 17,707,489.1 to 9,448,461.8, a relative improvement of 0.466 (95% interval $[0.268, 0.703]$). Median absolute error falls from 642.50 to 482.76 square feet. The log-scale comparison gives $\widehat{\lambda} = -0.038$, so the improvement in levels does not imply improvement under every error metric.

These results provide an independent reference for checking recorded physical inputs. Verification can draw on information produced outside the process through which officers maintain the register. The next question is whether that information also helps the authority decide which properties to inspect.

9.2 Targeting Verification

Inspection capacity is limited. Independent measurement can make that capacity more productive if discrepancies identify properties where a field visit is likely to reveal a recording error. Figure 9.1 evaluates a rule that ranks properties by the absolute log difference between administrative and remote-sensing measurements. Field measurements are then used to evaluate the selected properties; they do not enter the ranking. The log-based scores require positive measurements in all three sources, giving 2,796 land properties and 2,690 covered-area properties.

Among the 10% of properties with the largest land discrepancies, 76.3% have a field-confirmed administrative error, compared with 39.1% under random inspection. Targeting raises the error-detection rate by 37.3 percentage points, equivalent to approximately 37 additional errors identified per 100 inspections. Here, a field-confirmed error is an absolute administrative–field log difference exceeding 0.25.

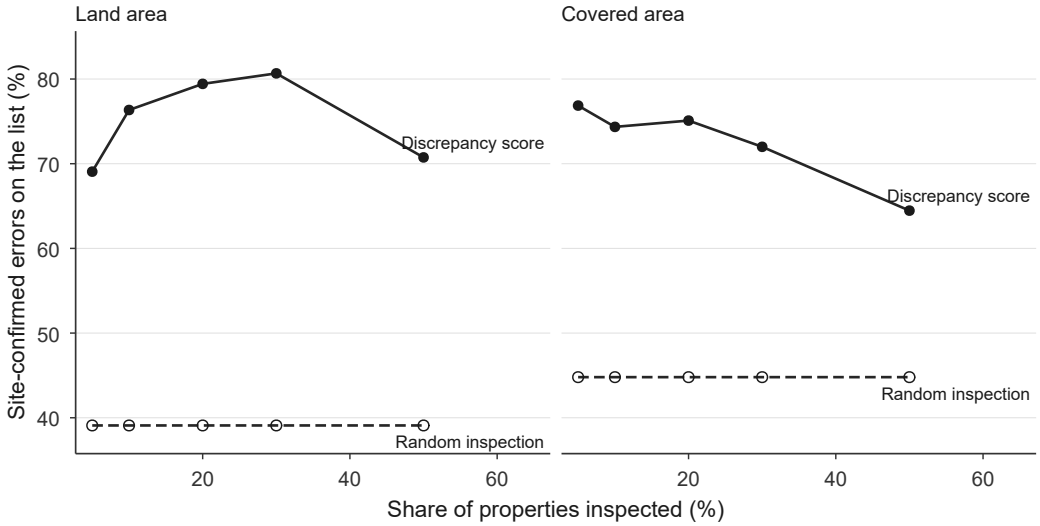


Figure 9.1: Targeting verification using independent measurement.

Notes: Properties are ranked by absolute administrative–remote-sensing log differences for the corresponding attribute. Precision is the share of inspected properties with an absolute administrative–field log difference exceeding 0.25. Field measurements do not enter the ranking score. The dashed line reports precision under random inspection.

The top-decile land list identifies 19.5% of all field-confirmed land errors and captures 39.2% of total absolute land error measured in logs. Thus, inspecting one tenth of properties reaches almost one fifth of errors and approximately two fifths of their total magnitude. For covered area, precision rises from 44.8% under random inspection to 74.3% under targeted inspection, a gain of 29.6 percentage points. Tables G.7 and G.8 report results across inspection shares and error tolerances.

These results give independent measurement a concrete role in supervising assessment. Discrepancy rankings allow the principal to direct verification towards records with a higher probability of error and obtain more findings from a given inspection budget. They also provide a basis for setting verification priorities alongside the collection targets faced by frontline officers.

9.3 Replacing Recorded Inputs Under a Common Schedule

The distributional consequences of record maintenance depend on how physical inputs translate into liabilities. I therefore apply the same capital-value schedule to alternative measurements of the same properties. Formula value combines land and structure values using the supplied District Collector land rates and structure rates. The common full-value schedule is

$$T(V) = \begin{cases} 0, & V \leq 5{,}000{,}000, \\ 0.0007V, & 5{,}000{,}000 < V \leq 10{,}000{,}000, \\ 0.0008V, & 10{,}000{,}000 < V \leq 25{,}000{,}000, \\ 0.0009V, & V > 25{,}000{,}000. \end{cases}$$

Rates apply to the full value within each band, creating notches at the thresholds.

The main comparison uses 7,956 properties in 65 assessment circles. I replace recorded land alone, covered area alone, and both inputs with their remote-sensing counterparts. Properties, official rates and supplied weights remain fixed across scenarios. All scenarios use the same ranking by remote-sensing formula value.

The weights adjust for observed differences between the measurement sample and the residential register. They use predicted sample-inclusion probabilities based on cadastral land and covered area, rental value, storeys, organised/planned/gated status and missing-data indicators. Predictions are generated across five groups of circles; inverse-probability weights are capped at the 99.5th percentile and normalised to the residential target file. These weights remain fixed throughout the comparisons and bootstrap draws.

For each replacement, I report mean liability, gross increases, gross reductions and the net change. Gross increases and reductions are averaged over all properties, including zeros where the bill does not change in the relevant direction. Progressivity is measured by the Kakwani index, $K = C_T - G_V$, where C_T is the concentration coefficient of tax liabilities and G_V is the Gini coefficient of formula value (Kakwani, 1977). A higher index means that higher-value properties bear a larger share of liabilities relative to their share of formula value. Uncertainty is calculated using 999 joint resamples of assessment circles.

9.4 Liability Changes and Progressivity

Table 9.2 reports the consequences of replacing physical inputs. Baseline mean liability is PKR 11,118.37. Replacing land alone reduces it by PKR 677.97, or 6.10%; replacing covered area alone increases it by PKR 899.77, or 8.09%. Joint replacement increases mean liability by PKR 133.78, or 1.20%, with a 95% interval of [PKR −819.68, PKR 1,223.56]. Because input changes can move properties across the schedule's notches, the separate-input effects need not sum to the joint effect.

Table 9.2: Fiscal and distributional consequences of replacing physical inputs

	Land only	Covered only	Both inputs
Mean liability (PKR)	10,440	12,018	11,252
	(1,291)	(1,427)	(1,333)
Mean net change (PKR)	−678	900	134
	(354)	(329)	(522)
Aggregate change (%)	−6.10	8.09	1.20
	(2.83)	(3.39)	(4.82)
Gross increases (PKR)	1,501	1,481	2,519
	(262)	(278)	(390)
Gross reductions (PKR)	2,179	581	2,386
	(357)	(74)	(379)
Kakwani index	0.0844	0.0474	0.0897
	(0.0167)	(0.0166)	(0.0107)
Change in Kakwani index	0.0423	0.0052	0.0476
	(0.0164)	(0.0126)	(0.0196)

Notes: All columns use the same 7,956 properties in 65 circles, supplied weights, official rates and RS formula-value ranks. Baseline mean liability is PKR 11,118.37 and the baseline Kakwani index is 0.0421. Parentheses contain standard deviations from 999 joint circle resamples. Joint-input percentile intervals are [PKR −819.68, PKR 1,223.56] for the net change and [0.0079, 0.0852] for the Kakwani change. Gross increases and reductions are averages across all properties. Rounded entries may not subtract exactly.

The modest net change combines substantial movements in opposite directions. Under joint replacement, gross increases average PKR 2,519.38 and gross reductions PKR 2,385.60 per property. These offsetting changes make aggregate liability an incomplete description of the consequences of reassessment.

Figure 9.2 restores the distributional comparison. Panel A plots the value Lorenz curve and the concentration curves for administrative and remote-sensing liabilities. Panel B shows the difference between each tax curve and the value curve. Joint replacement increases the Kakwani index from 0.0421 to 0.0897, a change of 0.0476 with a 95% interval of [0.0079, 0.0852]. Liabilities consequently become more concentrated on higher-value properties relative to their share of formula value.

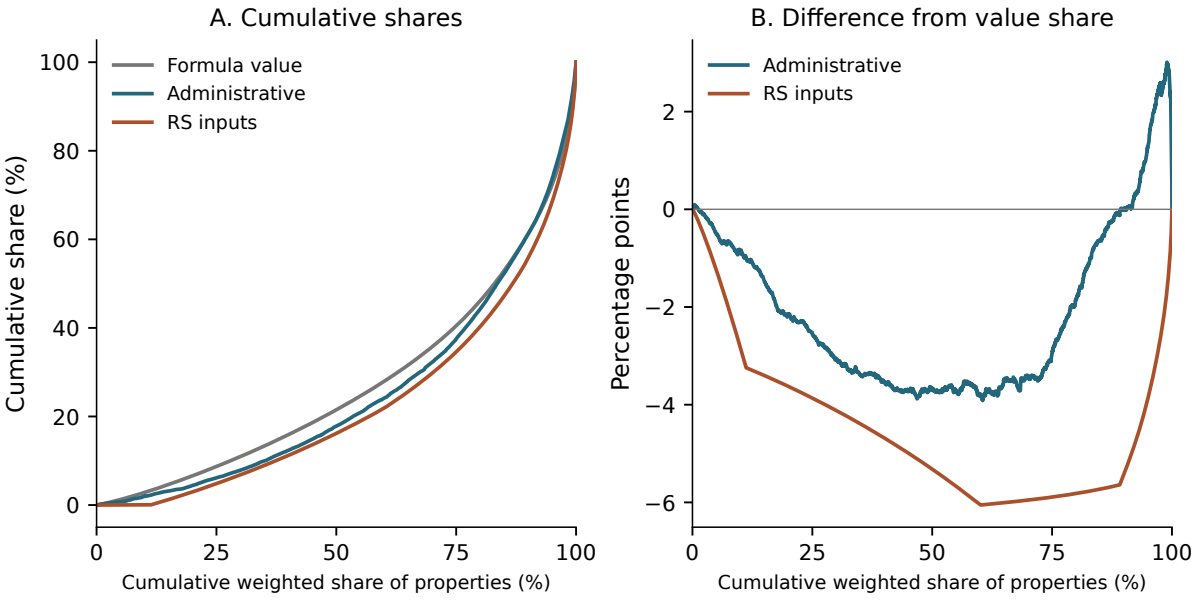


Figure 9.2: How replacing recorded inputs changes bills and their distribution.

Notes: Panels A and B rank properties from lowest to highest remote-sensing formula value. In Panel A, the horizontal axis is the cumulative weighted share of properties; each curve gives their cumulative share of property value or tax. A tax curve below the value curve means that properties below a given value rank account for a smaller share of tax than of property value. Panel B subtracts the value curve from each tax curve. All panels use the same 7,956 properties and weights.

The land-only comparison makes the distinction between revenue and distribution particularly clear. Replacing land increases the Kakwani index by 0.0423 while reducing aggregate assessed liability by 6.10%. More progressive liabilities can therefore accompany a smaller total assessed bill. The allocation of the burden depends on which properties receive increases and reductions, as well as their aggregate balance.

Replacing both recorded inputs with field measurements increases the Kakwani index by 0.0490, with a 95% interval of $[0.0117, 0.0919]$. Replacing land alone increases it by 0.0655, with an interval of $[0.0337, 0.1042]$. Joint field replacement raises mean assessed liability by PKR 871.78, or 5.35%, with a 95% interval of [PKR −435.89, PKR 2,282.85].

On the same sample and field-value ranking, joint remote-sensing replacement increases the Kakwani index by 0.0376, with a 95% interval of $[-0.0096, 0.0856]$. Appendix Table G.5 reports all replacement scenarios. The field-input results reinforce the central distributional finding: changing recorded characteristics changes the allocation of liabilities under an unchanged statutory schedule.

Taken together, the results connect information, supervision and tax distribution. Independent land measurement improves the reference available to the principal, and discrepancy rankings make inspections more productive. Applying the same schedule to alternative inputs also shows that maintaining records is consequential for who bears the tax. Technology can therefore support both verification and the implementation of the statutory allocation, giving the authority a basis for monitoring assessment alongside revenue collection.

10 Discussion: Delegation, Incentives and Distribution

10.1 The agency problem in maintaining the tax base

A statutory tax schedule allocates liabilities through the information contained in administrative records. Delegating the production and maintenance of that information gives frontline officers an important role in determining how the schedule operates. A reform can make the calculation more transparent while leaving the accuracy of its inputs dependent on officers' decisions about verification and amendment. The historical exemption patterns and persistence results show why this distinction matters: inherited entries can continue to shape liabilities after the assessment rule changes.

The principal–agent tension concerns how officers pursue the revenue mandate while valuing private returns and bearing the costs of administrative work. Senior officials describe increasing revenue as a central objective of reform, while frontline officers report anxiety about higher collection targets. Officers must therefore decide how to pursue receipts from a base whose maintenance also requires their effort. Recovering existing demand and correcting an understated assessment can both contribute to revenue, but they differ in the work involved, the additional demand created and the private savings preserved.

The private-return evidence connects progressive preferred burdens with the incentives governing implementation. Officers' preferred average tax rates rise with predicted market value. When correcting an understated assessment brings the bill closer to the officer's preferred amount, it raises the payoff from correction. Yet a sufficiently large tax saving can finance a payment that compensates the officer for retaining the lower bill. The citizen exercise shows that selected hypothetical concessions allocate part of the saving to assessors and that higher-value households accept the offered paid concessions more often. In the model, paid retention remains possible when $0 < F_{ij} < g_i$; correction proceeds once the officer's correction payoff reaches the taxpayer's saving. Progressive preferences can therefore coexist with persistent lower assessments when private compensation outweighs the gain from correction.

The second mechanism concerns the relative valuation of administrative outcomes. At equal collections and public support, 60.2% of fitted officer choices favour less delinquency despite greater underassessment under the question of which outcome will function effectively. Within the model, this preference makes recovery of existing demand attractive relative to work that improves assessment but creates further liabilities to pursue. Together, the two motives explain why accurate maintenance can receive less priority even when officers face pressure to raise revenue.

10.2 Why delegated assessment has distributional consequences

The public-finance significance of this agency problem extends to the allocation of the tax burden. A favourable record determines both the liability avoided by the taxpayer and the potential surplus available for a private arrangement. When favourable recording varies across the property-value distribution, administrative implementation changes the distribution delivered by the statutory schedule.

The input-replacement results demonstrate the importance of this channel. Holding the schedule fixed, joint remote-sensing replacement raises the Kakwani index from 0.0421 to 0.0897 using the formula-value ranking. The separate concentration comparison groups properties by predicted market value: those above PKR 25 million account for 18.0% of the weighted

population but 40.1% of positive potential savings. The field-input comparison also produces a more progressive allocation, with joint replacement increasing the index by 0.0490. These results show that the physical characteristics entered into the register are consequential for the relative burdens placed on lower- and higher-value properties.

Aggregate liability alone obscures much of this change. Under joint remote-sensing replacement, average increases of PKR 2,519.38 and reductions of PKR 2,385.60 largely offset, leaving a net increase of 1.20%. Land replacement alone improves measured progressivity while reducing aggregate assessed liability. Assessment accuracy, distribution and collection are therefore distinct dimensions of administrative performance. Evaluating implementation through receipts alone leaves the allocation produced by the underlying records insufficiently examined.

10.3 Technology and the supervision of assessment

Independent measurement gives the principal additional information with which to supervise delegated assessment. Remote sensing reduces land-area mean-squared error by approximately 57% and covered-area mean-squared error by approximately 47%. Median absolute error also falls for both attributes. The covered-area result differs on the log scale, making the choice of measurement task and performance criterion consequential.

The targeting results show how this information can guide administrative effort. Among the tenth of properties with the largest administrative–remote-sensing land discrepancies, 76.3% have a field-confirmed error, compared with 39.1% under random inspection. The same list captures 39.2% of total absolute land error measured in logs. Independent measurement therefore provides both a reference for checking records and a practical means of concentrating verification where it is more productive.

These functions address different aspects of the agency problem. Where private returns favour retaining understated bills, independent evidence can make favourable entries easier to scrutinise. Where verification is costly, discrepancy rankings can increase the number of errors identified with a given inspection budget. The institutional task is to connect this information to responsibility for reviewing discrepancies and completing justified amendments. Technology can supply measurements and verification priorities, while officers undertake the administrative decisions and collection work that follow. Its contribution depends on making assessment quality an observable part of performance alongside revenue collection.

11 Conclusion

How does delegating assessment and collection to officers with private interests shape the tax base and the distribution of liabilities? Evidence from Lahore connects persistent record discrepancies with two motives relevant to their maintenance: the private value of lower official bills and officers' preferences over assessment and recovery work. Historical exemption patterns remain visible in the register, while unchanged records carry substantial discrepancies forward. The choice exercises give these patterns an economic interpretation through preferred tax burdens, the division of private concessions and the priority placed on recovery at equal collections.

The consequences extend beyond the amount assessed. Replacing recorded inputs under the same statutory schedule changes who bears the liability, with greater measured progressivity in both the main remote-sensing comparison and the field-input comparison. Substantial increases

and reductions can coexist with a modest aggregate change. The distribution delivered by a property tax therefore depends on the administration of its base as well as the rates written into law.

Independent measurement offers a concrete means of strengthening oversight. More accurate land measurements make recorded inputs easier to check, and discrepancy-based targeting substantially improves the productivity of inspections. The broader implication is that a more verifiable tax calculation must be accompanied by attention to the incentives and information governing its inputs. Maintaining the tax base is itself a distributional task within revenue administration.

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A Figures & Tables

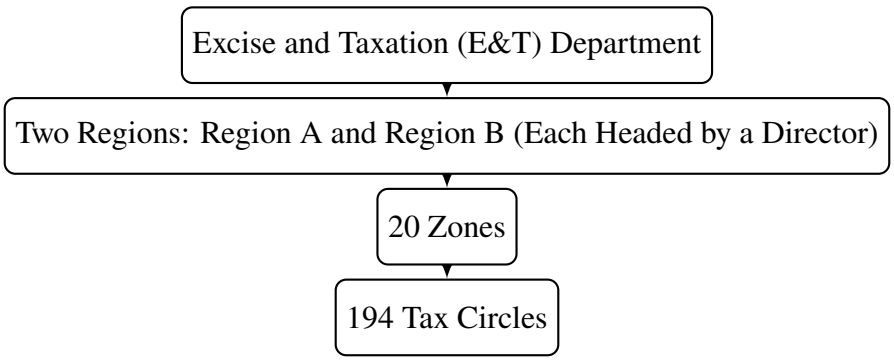


Figure A.1: Flowchart of Property Tax Administration in Lahore

RESIDENTIAL								
Category	Rented				Self			
	Rate of Land Sq. Yd. (In Rs.)		Rate of Covered Area in Sq. Ft (In Rs.)		Rate of Land Sq. Yd. (In Rs.)		Rate of Covered Area in Sq. Ft (In Rs.)	
	Upto 500	Exceeding 500	Upto 3000	Exceeding 3000	Upto 500	Exceeding 500	Upto 3000	Exceeding 3000
A	23	18.40	23	18.40	4.60	3.68	4.60	3.68
B	17	13.60	17	13.60	3.40	2.72	3.40	2.72
C	14	11.20	14	11.20	2.80	2.24	2.80	2.24
D	11	8.80	11	8.80	2.20	1.76	2.20	1.76
E	8.20	6.56	8.20	6.56	1.64	1.31	1.64	1.31
F	6.50	5.20	6.50	5.20	1.30	1.04	1.30	1.04
G	4	3.20	4	3.20	0.80	0.64	0.80	0.64

Figure A.2: Pre-reform valuation table used by Excise and Taxation frontline officers to calculate gross annual rental value and property-tax liability. Prescribed rates vary by location category, property use, and occupancy status; for otherwise comparable residential properties, the rate applied to rented property is five times the rate applied to self-occupied property.

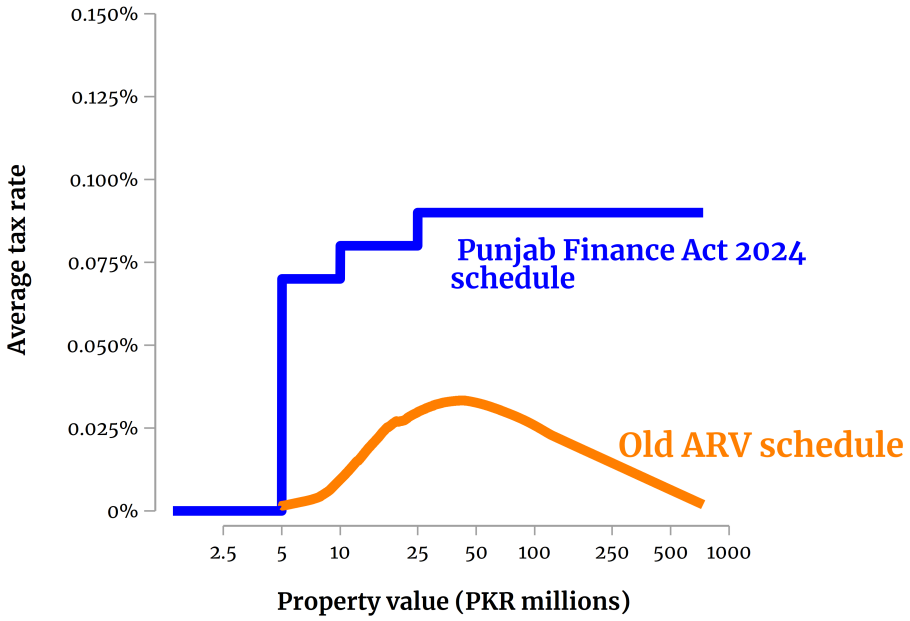
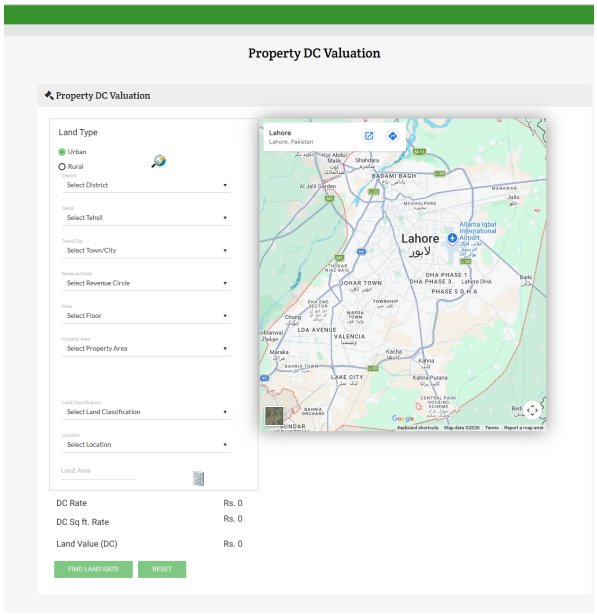


Figure A.3: Average property tax rates under the pre-reform ARV system and the Punjab Finance Act 2024 capital-value schedule. The figure compares the average tax rate implied by Punjab's pre-reform annual rental value (ARV) system with the capital-value-based schedule introduced by the Punjab Finance Act 2024. The average tax rate is defined as annual property tax liability divided by the property's capital value. Property values on the horizontal axis are estimated using the official District Collector (DC) valuation rates for 2021–22. The orange line shows the average tax rate under the old ARV system. Under this system, tax liabilities were determined from administratively assessed annual rental values rather than directly from the capital value of the property; the figure expresses these liabilities as a share of DC-rate property values to make the two systems comparable. The blue line shows the statutory schedule introduced by the Punjab Finance Act 2024, under which property tax is instead determined directly from capital value. Properties valued at up to approximately £13,400 (PKR 5 million) are exempt; properties valued above £13,400 and up to approximately £26,800 (PKR 10 million) are taxed at 0.07% of value; properties above £26,800 and up to approximately £67,000 (PKR 25 million) at 0.08%; and properties above £67,000 at 0.09%. Thus, conditional on crossing the exemption threshold, the statutory rates are low and increase only slightly across the three taxable value bands, producing a schedule that is nearly flat in average-rate terms. In contrast, the old ARV system generated a hump-shaped relationship between effective tax rates and capital values, with the tax burden rising among lower- and middle-value properties before declining substantially at higher property values. Sources: Punjab Finance Act 2024 for the capital-value schedule; Punjab Urban Immovable Property Tax (UIPT) MIS, 2021, for pre-reform ARV tax liabilities; Punjab DC valuation tables, 2021–22, for property values. GBP equivalents use an exchange rate of approximately £1 = PKR 373.3.



(a) Online DC valuation portal

VALUATION TABLE URBAN WAGHA TOWN

#	AREA	Residential 2018-19		Commercial 2018-19	
		Open Plot Per Marla	Constructed property per sqft	Open Plot Per Marla	Constructed property per sqft
1	ATTOKI AWAN, Bismillah , Al Raheem Garden , Al Ahmed Garden etc (All Residential)	275,000	880	375,000	1,430
2	BAGHBANPURA (ALL TOWN / SOCITIES)	375,000	880	700,000	1,430
3	BAGRIAN SYEDAN (ALL TOWN / SOCITIES)	250,000	880	500,000	1,430
4	CHAK RAMPURA (Garision Garden, Rehmat Town etc) (All Residential)	275,000	880	400,000	1,430
5	CHAK DHEERA (ALL TOWN / SOCIETIES)	400,000	880	1,000,000	1,430
6	DAROGHAWALA CHOWK TO RING ROAD MEHMOOD BOOTI	500,000	880	750,000	1,430
7	DAVI PURA (ALL TOWN / SOCITIES)	275,000	880	350,000	1,430
8	FATEH JANG SINGH WALA (ALL TOWN / SOCITIES)	400,000	880	1,000,000	1,430
9	GOBIND PURA (ALL TOWNS / SOCIEITIES)	400,000	880	1,000,000	1,430
10	HANDU, Al Raheem, Masha Allah, Gulshen Dawood,Al Ahmed Garden (ALL TOWN / SOCITIES)	250,000	880	350,000	1,430
11	JALLO, Al Hafeez, IBL Homes, Palm Villas, Aziz Garden etc	250,000	880	500,000	1,430
12	KHEERA, Aziz Garden, Canal Forts, Al Hafeez Garden, Palm Villas (ALL TOWN / SOCITIES)	250,000	880	500,000	1,430
13	KOT DUNI CHAND Al Karim Garden, Malik Nazir G Garden, Ghous Garden (ALL TOWN / SOCITIES)	250,000	880	400,000	1,430
14	KOTLI GHASI Hanif Park, Garision Garden, Gulshen e Haider, Moeez Town & New Bilal Gung H Scheme (ALL TOWN / SOCITIES)	250,000	880	500,000	1,430
15	LAKHODAIR, Al Wadood Garden (ALL TOWN / SOCITIES)	225,000	880	500,000	1,430
16	LAKHODAIR, Ring Road Par (ALL TOWN / SOCITIES)	75,000	880	200,000	1,430
17	LIDHAR (ALL TOWN / SOCITIES)	400,000	880	1,000,000	1,430
18	LEHNA SINGH WALA (ALL TOWNS / SOCIETIES)	400,000	880	1,000,000	1,430
19	MADSUDAN PURA Sharif Garden etc (ALL TOWN / SOCITIES)	300,000	880	500,000	1,430
20	MAHMOOD BOOTI in side Ring Road & New Shalimar etc (ALL TOWN / SOCITIES)	350,000	880	500,000	1,430
21	MAHMOOD BOOTI Ring Road Par (ALL TOWN / SOCITIES)	75,000	880	200,000	1,430
22	MANAWAN, Ghous Garden, Al Hafeez, Ashraf Garden, Atta Town, Tufail Garden & Rizwan Garden (ALL TOWN /	250,000	880	700,000	1,430

(b) Underlying DC valuation table

Figure A.4: Publicly accessible District Collector property-valuation system. Panel a shows the publicly accessible online District Collector (DC) valuation portal. A user can select a property's administrative location, land classification, use, floor, and area to retrieve the applicable land rate and implied DC value. Panel b shows an excerpt from the underlying DC valuation list for urban property in Wagha Town, Lahore, which is also publicly available. The list assigns separate benchmark rates to open land, expressed per marla, and constructed property, expressed per square foot, and distinguishes between residential and commercial use. Together, the two panels show that both the interface used to retrieve a property's official valuation and the location-specific rate schedules on which that valuation is based are available to the public. DC values are determined from administratively specified geographic categories and observable property characteristics rather than from a property-specific estimate of market value. Panel b displays the valuation schedule for 2018–19.

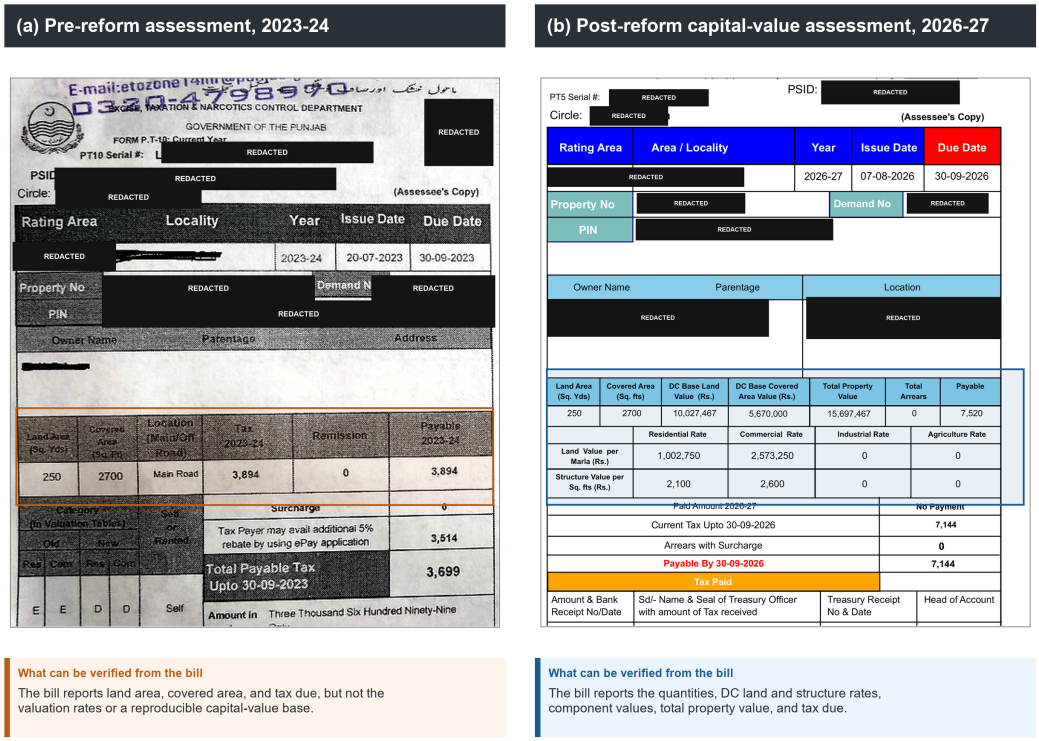


Figure A.5: **The capital-value reform made the assessment base reproducible from the tax bill.** Panel (a) shows the pre-reform Assessee's Copy for 2023–24. It reports the property's land area (250 square yards), covered area (2,700 square feet), and final tax demand, but it does not report the valuation rates applied to those quantities or the intermediate assessed property value. A taxpayer or supervisor therefore cannot reconstruct the tax base from the bill itself. Panel (b) shows the post-reform Assessee's Copy for 2026–27. The revised form reports the same physical quantities together with the District Collector (DC) rates—PKR 1,002,750 per marla of residential land and PKR 2,100 per square foot of residential structure—and separately displays the resulting land value (PKR 10,027,467), structure value (PKR 5,670,000), and total property value (PKR 15,697,467). The component values sum to the displayed total, and the published statutory tax schedule can then be applied to check the reported liability. The reform therefore did not merely change the tax base; it made the mapping from recorded property attributes to assessed value observable and externally verifiable. Direct taxpayer, property, account, location, QR-code, and officer identifiers have been redacted.

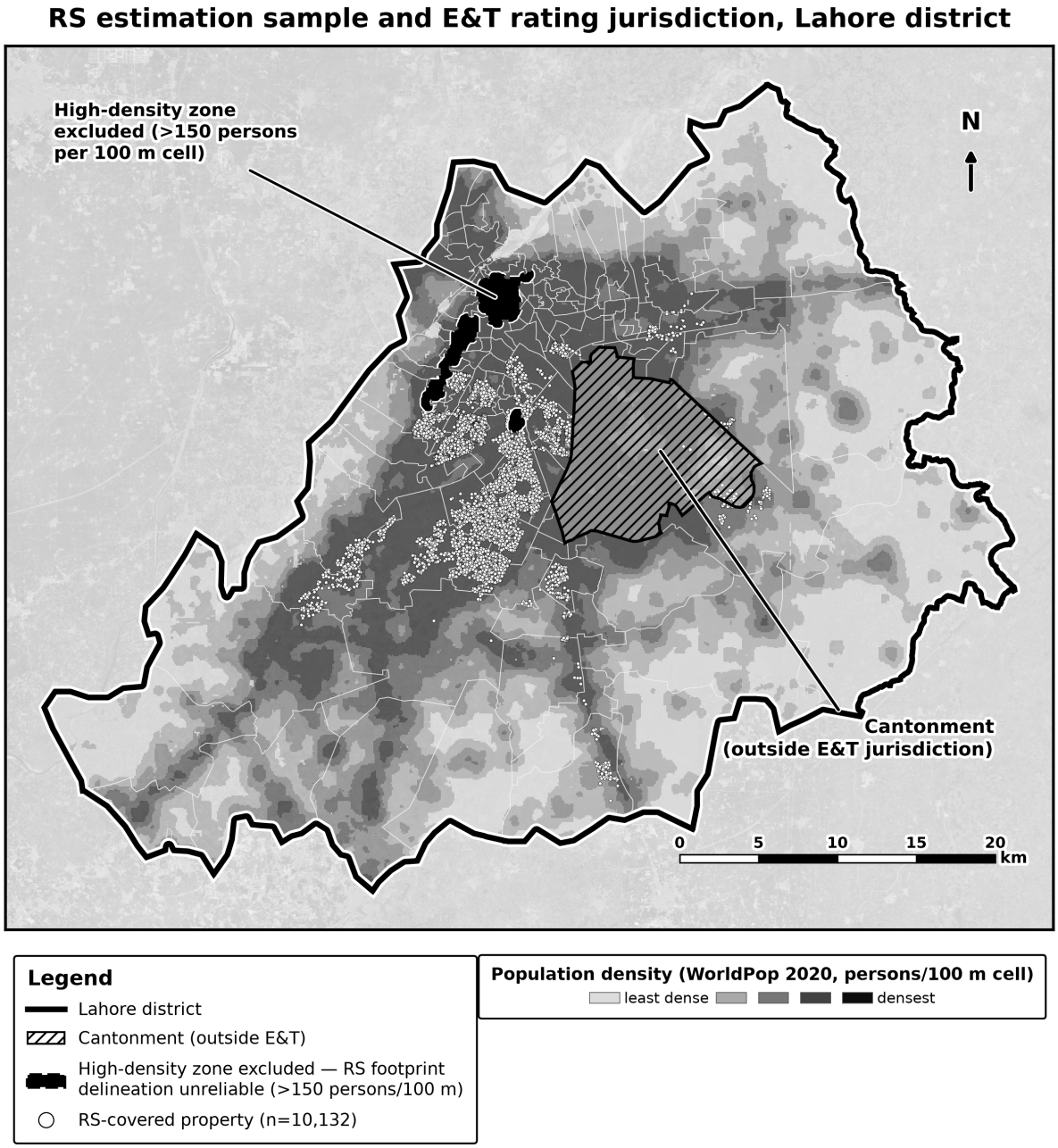


Figure A.6: The remote-sensing estimation sample within Lahore district. Grayscale shading is the WorldPop 2020 population surface (denser cells darker); thin lines are E&T rating jurisdiction. The Lahore Cantonment (hatched) is rated by the Cantonment Board and lies outside E&T jurisdiction; the dashed zone marks the high-density old-city fabric (above 150 persons per 100 m cell) where footprint delineation from imagery is unreliable. Open circles are the n = 10,132 RS-covered properties.

B Supporting model derivations

Fix the inherited assessment $v \equiv \bar{A}_i > 0$, with $0 \le d \le v$, and write $H = \gamma + \delta\bar{p}/v^2 > 0$. The payoff relative to zero departure is

$$\Pi(d; v) - \Pi(0; v) = (\theta - \mu)\{T(v) - T(v - d)\} - Hd^2.$$

If $\theta \le \mu$, a positive departure is strictly inferior to zero. Within an affine tax segment of slope τ, differentiation gives $(\theta - \mu)\tau - 2Hd = 0$ and second derivative $-2H < 0$. When the whole feasible interval is affine, the optimum is the interior expression clipped to $[0, v]$; otherwise segment-feasible candidates and boundary payoffs are compared. At $v = 0$, only zero departure is feasible.

The full-value schedule in Section 9.3 has notches. Its weak upper-band inequalities make $T(V)$ left-continuous at each cutoff, with an upward jump immediately to the right. For $\theta > \mu$, the departure objective is upper-semicontinuous on the compact feasible interval: landing exactly at a lower band's upper limit can dominate an interior first-order solution. Compare all feasible segment candidates with cutoff and endpoint payoffs. For $\theta \le \mu$, zero departure is optimal. The convergence bound below requires only the stated bounded monotone schedule and attained optima; it is not restricted to a smooth schedule. Threshold incentives can favour boundary choices, but the historical recording design is not a structural estimate of this departure problem.

For $\gamma_2 > \gamma_1$, let d_2, d_1 be attained optima. Adding their optimality inequalities yields $(\gamma_2 - \gamma_1)(d_1^2 - d_2^2) \ge 0$, hence $d_2 \le d_1$. Comparing an optimum with zero also gives

$$0 \le (d^*)^2 \le \frac{(\theta - \mu)_+\{T(v) - T(0)\}}{\gamma}.$$

Thus departures tend to zero as checkability increases, holding the record fixed. This is convergence toward the inherited benchmark; it need not correct its physical inputs.

B.1 Assessed-liability credit as a special case

Restore the officer index on the revenue weight: μ_j is the same kind of weight as μ in the original assessment problem. At $d = 0$, that problem contributes only revenue credit: $\mu_j T_i^0$ under retention or $\mu_j T_i^1$ under correction. There is no departure, departure-related private saving or departure cost. Adding the record return, the preferred-bill term and amendment cost gives

$$J_{ij}(0) = \mu_j T_i^0 + \theta_j^x [g_i]_+ - \lambda_j |T_i^0 - P_{ij}|, \tag{B.1}$$

$$J_{ij}(1) = \mu_j T_i^1 - \lambda_j |T_i^1 - P_{ij}| - c_{ij}. \tag{B.2}$$

Retention earns revenue credit from the inherited bill and can preserve a private return. Correction earns credit from the corrected bill, removes that record return and costs c_{ij}. Both actions are judged against the same preferred bill P_{ij}.

Subtracting the complete action payoffs therefore gives

$$\begin{aligned} J_{ij}(1) - J_{ij}(0) = {} & \underbrace{\mu_j g_i}_{\text{revenue credit}} - \underbrace{\theta_j^x [g_i]_+}_{\text{private return forgone}} \\ & + \underbrace{\lambda_j q_{ij}}_{\text{preferred-bill effect}} - \underbrace{c_{ij}}_{\text{amendment cost}}. \end{aligned} \tag{B.3}$$

Where amendment is feasible, the inspector corrects precisely when

$$\boxed{\mu_j g_i + \lambda_j q_{ij} > \theta_j^x [g_i]_+ + c_{ij}.} \tag{B.4}$$

The record is retained at equality or when correction is infeasible. A larger bill can increase revenue credit while taking the bill further from what the inspector considers appropriate. Removing a taxpayer saving can also remove a private return. Neither accurate calculation nor a positive fiscal gain therefore guarantees correction.

For illustration, suppose $T_i^0 = 5{,}000$, $T_i^1 = 10{,}000$ and $P_{ij} = 6{,}000$. Correction raises liability by $g_i = 5{,}000$, but changes distance from the preferred bill from 1,000 to 4,000, so $q_{ij} = -3{,}000$. Its payoff advantage is $5{,}000(\mu_j - \theta_j^x) - 3{,}000\lambda_j - c_{ij}$. These amounts illustrate the mechanism, not estimated parameters.

Correction comparative statics in the assessed-liability special case. For a feasible correction define its payoff advantage as

$$\Delta_{ij} = \mu_j g_i - \theta_j^x [g_i]_+ + \lambda_j q_{ij} - c_{ij}.$$

Its derivatives with respect to $\mu_j, \theta_j^x, \lambda_j, c_{ij}$ are respectively $g_i, -[g_i]_+, q_{ij}, -1$. Higher revenue credit favours upward correction and can discourage downward correction; a larger retention return discourages upward correction; lower amendment cost favours either direction. Stronger attachment to the preferred bill encourages correction precisely when $q_{ij} > 0$. These are conditional statements holding the property's other terms fixed.

An intermediate correction interval follows in a specific case. Let $g > 0$, $a = P - T^0 > 0$, and hold $a, c, \mu, \theta^x, \lambda$ fixed. Then $q = a - |g - a|$ and

$$\Delta(g) = \begin{cases} (\mu - \theta^x + \lambda)g - c, & 0 < g \le a, \\ (\mu - \theta^x - \lambda)g + 2\lambda a - c, & g > a. \end{cases}$$

If $-\lambda < \mu - \theta^x < \lambda$ and $(\mu - \theta^x + \lambda)a > c$, correction occurs only for

$$\frac{c}{\mu - \theta^x + \lambda} < g < \frac{2\lambda a - c}{\lambda - \mu + \theta^x}.$$

The interval illustrates how a larger fiscal gain can eventually move the bill too far from the preferred amount. Equality selects retention.

Revenue-led benchmark. With no preferred-bill motive and $\mu > \theta^x$, correction at a positive gain requires $(\mu - \theta^x)g > c$. For a fixed nonnegative cost distribution, its probability is zero at $g \le 0$ and weakly increasing at $g > 0$. This special-case restriction is stronger than the collection-target prediction in the main text; it does not prohibit downward corrections in the general target model.

Collection targets. Suppressing property and officer indices, let B be total demand and R expected receipts in the primary collection-target model. Let T be the focal property's assessed liability, with credit weight μ. When both actions leave a positive shortfall, the fiscal payoff is $\mu T + \eta R - a(B - R)$. Correction changes the property's bill and circle demand by g, and receipts by r, so its incremental fiscal return is $\mu g + (\eta + a)r - ag$. Adding career recognition κh, preferred-bill alignment λq, the forgone private return and amendment cost gives equation (4.11). Substituting $r = pg$ gives equation (4.13); its fiscal cross-partial with respect to p and g is $\eta + a \ge 0$.

C Negotiated private returns at two margins

An inspector can negotiate a payment for assessing below the formula value of a given record, or for leaving an understated record unchanged. The same bargaining method applies to both decisions. What changes is the taxpayer's saving and the value of the action the inspector gives up. This appendix derives the payments used in the main model and distinguishes them from the shares displayed in the citizen experiment.

C.1 The common bargaining calculation

Let $G > 0$ be the taxpayer's gross saving from an agreement and $K \geq 0$ the inspector's payoff loss from accepting it before any bribe is paid. Relative to disagreement, the taxpayer gains $G - b$ and the inspector gains $b - K$. The payment b transfers one rupee of payoff from one party to the other. Both can gain only if $G > K$. With officer bargaining weight $\beta^B \in (0, 1)$, the asymmetric Nash solution (Nash, 1950; Kalai, 1977) solves

$$b^* = \arg \max_{K \leq b \leq G} (b - K)^{\beta^B} (G - b)^{1-\beta^B}. \tag{C.5}$$

Taking logarithms gives the first-order condition

$$\frac{\beta^B}{b - K} - \frac{1 - \beta^B}{G - b} = 0.$$

The second derivative is $-\beta^B/(b - K)^2 - (1 - \beta^B)/(G - b)^2 < 0$. Thus the unique interior payment is

$$\begin{aligned} b^* &= K + \beta^B (G - K), \\ b^* - K &= \beta^B (G - K), \qquad G - b^* = (1 - \beta^B)(G - K). \end{aligned} \tag{C.6}$$

The inspector is compensated for the alternative forgone, then receives a fraction of the remaining gain. The resulting payment share is

$$\frac{b^*}{G} = \beta^B + (1 - \beta^B)\frac{K}{G}. \tag{C.7}$$

It equals the bargaining weight only when $K = 0$. Holding G and β^B fixed, raising K by one rupee raises the payment by $1 - \beta^B$. If $G \leq K$, there is no strictly beneficial agreement; I select disagreement at equality. The gain being divided is private to the two parties and does not count the loss of official revenue as a social benefit.

C.2 Underassessment and retention have different payoffs

For underassessment, disagreement means applying the formula to the existing record without a bribe. For retention, a paid agreement is possible when disagreement would lead the inspector to correct an understated record. The two savings and officer losses are

$$\begin{aligned} G_i^a(d_i) &= T_i(\bar{A}_i) - T_i(\bar{A}_i - d_i), \quad K_i^a(d_i) = \mu G_i^a(d_i) + H_i d_i^2, \\ G_i^x &= g_i > 0, \qquad\qquad K_{ij}^x = F_{ij} > 0. \end{aligned} \tag{C.8}$$

For positive joint gains, the payments are

$$\begin{aligned} b_i^{a*}(d_i) &= K_i^a(d_i) + \beta_a^B \{G_i^a(d_i) - K_i^a(d_i)\}, \\ b_{ij}^{x*} &= F_{ij} + \beta_{x,j}^B (g_i - F_{ij}). \end{aligned} \tag{C.9}$$

The bargaining weights may differ across the two negotiations. Even if they are equal, different losses relative to disagreement can produce different payment shares. The model holds the weights fixed in its comparative statics and allows both payments and shares to adjust.

In the assessment negotiation, the inspector's gain after payment is $\beta_a^B S_i^a(d_i)$ and the taxpayer's is $(1-\beta_a^B)S_i^a(d_i)$, where $S_i^a = (1-\mu)G_i^a - H_i d_i^2$. Choosing the departure jointly with the transfer therefore maximises S_i^a. Equivalently, an inspector anticipating the negotiated payment chooses that same departure. This yields equation (4.3), with boundary comparisons in Appendix B. The payment compensates lost liability credit and departure costs; those costs are not subtracted twice.

C.3 The correction option and participation

The record comparison uses the inspector's best action without a private payment as the consequence of failed negotiation. Before a bribe, the payoff from correction minus retention is, under positive shortfalls,

$$F_{ij} = \mu_j g_i + (\eta_j + a_j) r_{ij} - a_j g_i + \kappa_j h_{ij} + \lambda_j q_{ij} - c_{ij}. \tag{C.10}$$

If $g_i > 0$ and $F_{ij} > 0$, correction is the inspector's preferred unpaid action. The taxpayer's saving from preventing it is g_i, valued rupee for rupee, and the inspector gives up F_{ij}. When $0 < F_{ij} < g_i$, bargaining gives

$$\begin{aligned} b_{ij}^{x*} &= F_{ij} + \beta_{x,j}^B (g_i - F_{ij}), \\ b_{ij}^{x*} - F_{ij} &= \beta_{x,j}^B (g_i - F_{ij}) > 0, \\ g_i - b_{ij}^{x*} &= (1-\beta_{x,j}^B)(g_i - F_{ij}) > 0. \end{aligned} \tag{C.11}$$

The officer's correction advantage after the negotiated payment is $\beta_{x,j}^B (F_{ij} - g_i) < 0$, so the agreed outcome is retention. If $F_{ij} \geq g_i$, no transfer can give both a positive gain; correction follows, including at equality under the stated no-agreement convention. If $F_{ij} \leq 0$, the inspector would retain the record without payment, including at a tie. The taxpayer then has no reason to pay for retention, and $b_{ij}^x = 0$. The inspector cannot support a payment by threatening a correction that would not be carried out. If correction is infeasible, the same unpaid-retention outcome applies. For $g_i \leq 0$, there is no taxpayer saving from retention, and the model allows no bribe for facilitating correction; the inspector corrects exactly when $F_{ij} > 0$.

Within the paid-retention region, the payment share is $\theta_{ij}^x = \beta_{x,j}^B + (1-\beta_{x,j}^B) F_{ij}/g_i$. Its derivative with respect to F_{ij} at fixed g_i is $(1-\beta_{x,j}^B)/g_i > 0$. At fixed $F_{ij} > 0$, a larger saving raises the payment but lowers its share. When the correction payoff itself depends on the saving, the total payment derivative is

$$\frac{db_{ij}^{x*}}{dg_i} = \beta_{x,j}^B + (1-\beta_{x,j}^B)\frac{dF_{ij}}{dg_i}.$$

Changing bargaining power raises the payment by $g_i - F_{ij}$ per unit change in $\beta_{x,j}^B$, without changing the existence of a mutually beneficial agreement. These derivatives describe the interior region, not transitions between paid retention, unpaid retention and correction.

An equal-receipts task alternative. If recovery rather than inaction is the alternative to correction, its payoff must enter disagreement. Suppose recovery leaves the focal record unchanged,

gives the same expected receipts as correction, and adds no separate effort cost, recognition or preferred-bill change. Suppose also that the taxpayer's saving from retaining the focal record remains g_i. The unpaid payoff difference is then $D_{ij} = (\mu_j - a_j)g_i + \kappa_j h_{ij} + \lambda_j q_{ij} - c_{ij}$. Replace F_{ij} by D_{ij} throughout the retention bargain: for $0 < D_{ij} < g_i$, the payment is $D_{ij} + \beta^B_{x,j}(g_i - D_{ij})$; for $D_{ij} \geq g_i$, correction is chosen; and for $D_{ij} \leq 0$, recovery with the old record is chosen without a payment. Additional recovery costs or recognition must be included in D_{ij}. A payment derived against inaction cannot be held fixed when the alternative changes to recovery.

C.4 Displayed payment shares in the citizen experiment

For a selected hypothetical paid inspector offer, let $G > 0$ be the displayed reduction in the official bill and b the inspector payment. When $0 \leq b \leq G$, the monetary shares are

$$s^O = \frac{b}{G}, \qquad s^C = \frac{G - b}{G} = 1 - s^O. \tag{C.12}$$

The ratio $b/(G - b)$ instead compares the payment with the taxpayer's net saving. These are accounting measures. The bargaining model shows why a payment share can reflect both bargaining power and the officer's alternative; the citizen experiment presents offers and does not observe that negotiation.

In citizen BACE option (i, t, k), i identifies the respondent, t the intended task from 1 to 10, and k an alternative. Paid human-assessor options have `human=1`, `bri`> 0 and `reduce`> 0. Set $b_{itk} = \texttt{bri}_{itk}$ and $G_{itk} = \texttt{reduce}_{itk}$. The source identities are `fin=real` + `mistake` $- G$ and `oop=fin` $+ b$. For example, a PKR 5,800 reduction with a PKR 3,500 payment leaves the taxpayer PKR 2,300: the payment share is 60.34% and the taxpayer share is 39.66%. Neither number alone identifies a bargaining weight.

Let $\mathcal{K}_i^{\text{ch}}$ contain respondent i's chosen paid-deal options, n_i^{ch} their number, and $\mathcal{I}_c^{\text{ch}}$ the N_c^{ch} respondents in circle c who choose at least one. The reported circle statistic is

$$\bar{s}_c^O = \frac{1}{N_c^{\text{ch}}} \sum_{i \in \mathcal{I}_c^{\text{ch}}} \left[\frac{1}{n_i^{\text{ch}}} \sum_{(t,k) \in \mathcal{K}_i^{\text{ch}}} \frac{b_{itk}}{G_{itk}} \right], \qquad \bar{s}_c^C = 1 - \bar{s}_c^O. \tag{C.13}$$

Respondents receive equal weight regardless of the number of paid deals selected. Those choosing none do not enter this conditional mean and are not assigned a zero share. Matched offered means use the same respondents as the chosen-share calculation. Section 8 presents these conditional shares alongside the distribution of formula-implied savings and the separate acceptance comparison. The empirical shares retain their accounting meaning under the negotiated-payment model.

D Construction of remote-sensing-based covered floor area

This appendix describes the construction of a property-level proxy for total covered floor area. The measure combines remotely estimated building footprints and heights with the relationship between storey count and relative floor areas observed in administrative records. Administrative data supply the typical vertical floor-area profile; remote sensing supplies the scale to which that profile is applied. Consequently, the estimate does not substitute a property's recorded administrative covered area for its remotely estimated area.

D.1 Footprint and storey count

Let $\widehat{L}_i$ denote the remotely estimated plot area of property i, measured in square metres, and let $\widehat{c}_i$ denote its estimated covered percentage. The estimated building footprint, expressed in square feet, is

$$\widehat{B}_i = 10.7639\,\widehat{L}_i \frac{\widehat{c}_i}{100}. \tag{D.1}$$

The conversion assumes that plot area and coverage refer to the same property boundary. To translate footprint into total floor area, we treat the remotely estimated footprint as a proxy for the area of the largest floor. This correspondence is a measurement assumption: roof projections, overhangs and differences in floor layouts can cause the two areas to diverge.

Let $\widehat{H}_i$ denote predicted building height in metres. For properties with valid positive height predictions, the storey classification is

$$\widehat{S}_i = \max\left\{1, \operatorname{round}\left(\frac{3.28084\,\widehat{H}_i}{h_0}\right)\right\}, \qquad h_0 = 10, \tag{D.2}$$

where h_0 is the assumed effective height per storey in feet. This classification is an approximation based on height rather than a direct observation of floors. Its interpretation requires administrative storey counts and height-based counts to use a common convention, including the treatment of the ground floor.

D.2 Administrative floor-area profiles

For administrative property j, let a_{jf} denote recorded covered area on floor f. The construction uses the available ground-floor and first- through fifth-floor fields. Define

$$T_j = \sum_{f=0}^{5} a_{jf}, \qquad A_j = \max_{f\in\{0,\ldots,5\}} a_{jf}, \qquad R_j = \frac{T_j}{A_j}, \tag{D.3}$$

where T_j is total recorded floor area, A_j is the largest recorded floor area and R_j is the total-to-largest-floor ratio. The ratio is invariant to the common unit in which floor areas are measured. It measures effective floor-area coverage across storeys, rather than the number of storeys itself. For example, a building with a largest floor of 1,000 square feet and another floor of 600 square feet has a multiplier of 1.6 rather than 2.

The calibration sample retains administrative properties with recorded land area between 25 and 2,000 square yards, at least one nonmissing floor-area entry, and positive largest-floor area.

It additionally excludes observations with largest-floor area exceeding 160% of recorded land area after conversion to a common unit.[3]

Let S_j be the administrative storey count. We estimate a separate multiplier for each observed storey category using Poisson pseudo-maximum likelihood (PPML):

$$\mathbb{E}[T_j \mid A_j, S_j = s] = A_j \exp(\alpha_s) = A_j M_s. \tag{D.4}$$

The specification includes a saturated set of storey categories and an offset of $\log A_j$, fixing the elasticity with respect to largest-floor area at one. PPML is used as a conditional-mean estimator; it does not require covered area to be a count or to follow a Poisson distribution.

For the estimation sample $\mathcal{J}_s$ in category s, the first-order condition is

$$\sum_{j \in \mathcal{J}_s} \left(T_j - A_j \widehat{M}_s \right) = 0. \tag{D.5}$$

It follows that

$$\widehat{M}_s = \frac{\sum_{j \in \mathcal{J}_s} T_j}{\sum_{j \in \mathcal{J}_s} A_j} = \sum_{j \in \mathcal{J}_s} w_{js} R_j, \qquad w_{js} = \frac{A_j}{\sum_{k \in \mathcal{J}_s} A_k}. \tag{D.6}$$

The estimated multiplier is therefore the largest-floor-area-weighted mean of the individual ratios within a storey category. It reproduces total recorded area within that calibration category. This aggregation property holds in the estimation sample and does not, by itself, establish accuracy in the remote-sensing sample.

D.3 Predicted total covered area

The calibrated estimate is

$$\boxed{\widehat{T}_i^{\mathrm{RS}} = \widehat{B}_i \widehat{M}_{\widehat{S}_i}.} \tag{D.7}$$

Thus properties assigned to the same storey category receive the same relative floor-area multiplier, but their predicted total areas vary with their remotely estimated footprints. For example, a multiplier of 1.6 applied to a footprint of 1,200 square feet yields a total covered-area estimate of 1,920 square feet. A multiplier is not available for storey categories absent from the administrative calibration sample.

The corresponding equal-floor benchmark is

$$\widehat{T}_i^{\mathrm{equal}} = \widehat{B}_i \widehat{S}_i. \tag{D.8}$$

This benchmark assumes that each storey occupies the entire footprint. The calibrated measure instead incorporates the relative floor-area patterns found in administrative buildings of the same storey category.

The construction can also be expressed as a sum of floor-specific contributions. On a common calibration sample, define

$$m_{fs} = \frac{\sum_{j \in \mathcal{J}_s} a_{jf}}{\sum_{j \in \mathcal{J}_s} A_j}. \tag{D.9}$$

[3]The implementation treats administrative floor areas as square feet and converts land area from square yards by multiplying by nine. Missing floor entries do not contribute to the sum, and the maximum is taken over nonmissing entries. Thus interpretation as total area requires missing entries to represent absent floors rather than unrecorded existing floors, and requires the available floor fields to cover the building. The coverage restriction is a data-cleaning rule; it does not establish that the retained observations are correctly measured.

Then $\sum_f m_{fs} = \widehat{M}_s$ and $\widehat{T}_i^{\mathrm{RS}} = \sum_f \widehat{B}_i m_{f,\widehat{S}_i}$. This identity clarifies how the multiplier incorporates relative floor coverage. The implemented estimator constructs the summed multiplier directly and produces total covered area, rather than separate predictions for each floor.

D.4 Measurement assumptions and interpretation

The central transfer assumption is that the relative floor-area profiles in the administrative calibration sample are informative about the target properties conditional on storey count. In conditional-mean terms, applying equation (D.7) to true covered area requires

$$\mathbb{E}[T_i^* \mid B_i^*, S_i = s,\ i \in \mathcal{P}] = B_i^* M_s, \tag{D.10}$$

where $\mathcal{P}$ denotes the target population and stars denote true quantities. The administrative calibration must identify the relevant M_s, while the remote-sensing footprint and storey classification must adequately approximate their true counterparts. Differences in building use, plot size or construction patterns within storey categories can weaken this transfer. The administrative plot-size restrictions also limit the population from which the profiles are learned.

Using relative profiles reduces dependence on administrative area levels. To see this, suppose all floors within property j share a positive reporting factor q_j, so that

$$a_{jf} = q_j a_{jf}^* \quad \text{for every } f. \tag{D.11}$$

Then

$$R_j = \frac{q_j T_j^*}{q_j A_j^*} = \frac{T_j^*}{A_j^*}. \tag{D.12}$$

Proportional understatement across all floors therefore leaves the property's ratio unchanged. This invariance does not extend to selective omission of floors or different reporting factors across floors. Moreover, the pooled multiplier remains weighted by recorded largest-floor area:

$$\widehat{M}_s = \frac{\sum_{j \in \mathcal{J}_s} q_j T_j^*}{\sum_{j \in \mathcal{J}_s} q_j A_j^*}. \tag{D.13}$$

Variation in reporting factors across properties can consequently change the pooled multiplier by changing the weights, even when each individual ratio is preserved. The method therefore borrows relative administrative patterns without being wholly independent of administrative reporting quality.

Finally, the constructed measure is a proxy for total covered floor area. Its errors can arise from remote-sensing inputs, height-based storey classification, mismatches between buildings and taxable properties, changes between observation dates, and imperfect transfer of administrative floor profiles. In the absence of independently measured floor areas, differences between this proxy and recorded administrative totals cannot uniquely identify underreporting. Comparisons with administrative records assess consistency with those records rather than independently validating true covered area.

E Property Value Prediction

This appendix describes the procedures we followed to estimate predicted property values for all 802,000 properties in the Excise & Taxation cadastre. Section E.1 describes our survey with real estate agents to create the training data used for our estimation. Section E.2 describes how we impute localities for the parts of the cadastre missing locality information. Section E.3 describes the random forest algorithm and its performance.

E.1 Training Data: Real Estate Agent Valuation

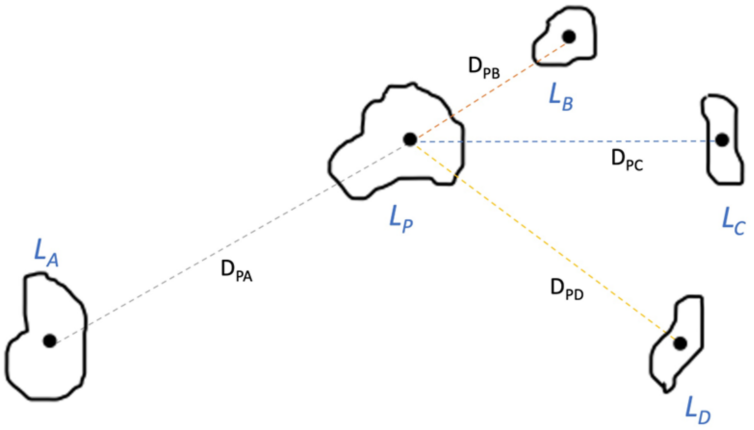
We get 2023 market value data from real-estate experts. For every property assessed, participating dealers were requested to provide estimates on the capital value, potential value as an open plot, and the rental value. Dealers were also asked about their own confidence levels in the reported values and their observations regarding property trends over the past six months, as well as their expectations for the next six months.

E.2 Imputing Missing Locality Data

One of the key inputs into our property valuation algorithm is the neighborhood a property is located in. However, the Excise & Taxation cadastre only contains the names of localities, not geocoded data on their location. To impute this missing data we follow the following procedue. Localities were categorized into one of four distinct groups.

- **Type I (TI):** This category included localities that were present in the valuation sample. Property geocodes obtained during the valuation exercise were utilized for these localities to determine a quasi-centroid in cases where the cadastre lacks geocodes. Once the quasi-centroids were obtained for all TI localities, the localities where the rates were missing were assigned DC residential and commercial rates from nearest possible locality using Mahalanobis distances

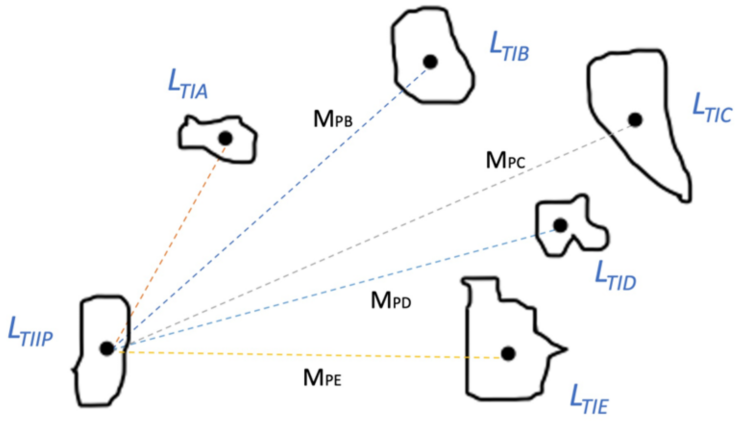
Figure E.1: Assigning DC values to a TI locality using location attributes



Source: IDEAS-LUMS Property Valuation Survey 2023
Notes: Figure shows if rates were missing from locality L_p, commercial and residential locality rates were assigned from locality L_B as its nearest to L_P. The distance was determined using Mahalanobis distance.

- **Type II (TII):** This category included localities which were not drawn in the main sample but had geo-codes and commercial and residential locality-level rates from the DC 2018-19 list. For each TII locality, Mahalanobis distances were computed for their proximity with all TI localities using location (i.e. longitude and latitude) and fanciness (i.e. DC residential and commercial rates) attributes. They were then linked to the closest TI locality (see Figure E.2) for the prediction model.

Figure E.2: Linking a TII locality to a TI locality using location and fanciness attributes

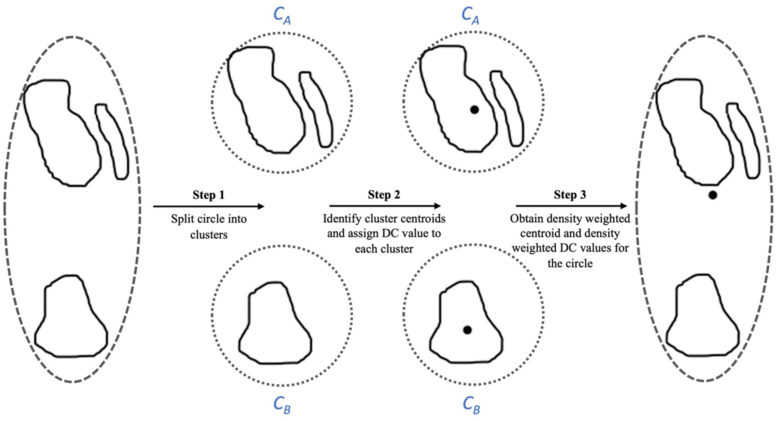


Source: IDEAS-LUMS Property Valuation Survey 2023
Notes: Figure shows out-of-sample L_{TIIP} locality was matched to L_{TIA} due to its proximity in terms of Mahalanobis distance.

- **Type III (TIII):** These types of localities were not drawn into the sample but had location geo-codes, and at least one of the residential and commercial DC rates was missing. For each TIII locality Mahalanobis distances were computed to all TI and TII localities using location attributes (i.e. longitude and latitude). Each TIII locality was then assigned DC rates of the closest TI or TII locality. If the TIII locality was assigned DC rates of a TI locality, then it was also linked to the same TI locality for the prediction model. Otherwise, location and "estimated" fanciness measures were used to link this TIII locality to a TI locality.

- **Type IV (TIV):** These localities were not drawn in the sample and did not have geocoded location information or DC rates. TIV localities were first split into two subtypes: a) locality lies in E&T defined circle that has geocodes and DC rates; and b) locality lies in a E&T defined circle that has no geocodes property and no DC rates. For a), missing information was filled using the strategy employed for TII localities. The only change was that circle centroids and average DC rates at the circle level were assigned to type a) localities. For localities from sub-sample b), 37 circle boundaries were plotted on QGIS. 14 of these circles were scattered around different parts of the city. It was decided that these circles would be (manually) split into clusters and cluster centroids and densities were computed using AsiaPop data. Each cluster was assigned DC rates of the closest TI, TII or TIII locality. These values were then computed using a density-weighted centroid and density-weighted DC residential and commercial rates for each circle (see Figure E.3). Mahalanobis distances were computed in the final step to link each type b) locality to a TI locality using density-weighted centroid and density-weighted DC rates.

Figure E.3: Dealing with a circle that has clusters in different parts of the city



Source: IDEAS-LUMS Property Valuation Survey 2023

E.3 Random forest property value data

This section details the procedures adopted to generate the random forest data. One source of estimating baseline levels of progressivity is administrative data. For this purpose, rental and capital market values for 2023 were obtained by surveying a sample of 12,363 commercial and residential properties from real estate experts. This 12,363 sample was then expanded to 802,592 properties using random forest. The random forest data was then merged with the property tax collection data obtained from E&T to create a unique dataset that contains information on 2022-2023 capital and rental market values and actual tax liabilities from FY 2021-2022.

The E&T property cadastre has 2,069 localities, of which only 407 were sampled. Predicting the property values for 1,662 out-of-sample localities was crucial because the neighbourhood is one of the key determinants of property value.

For this purpose, the location and average value of the locality were used to predict the property values. For location, an average of the property geocodes in the locality was taken to get a quasi-centroid (with latitude and longitude). Secondly, DC 2018-19 land rates (and not structure rates) served as a measure of fanciness for that locality. Both residential and commercial DC land rates were used to link the localities as they significantly differ even within a locality.

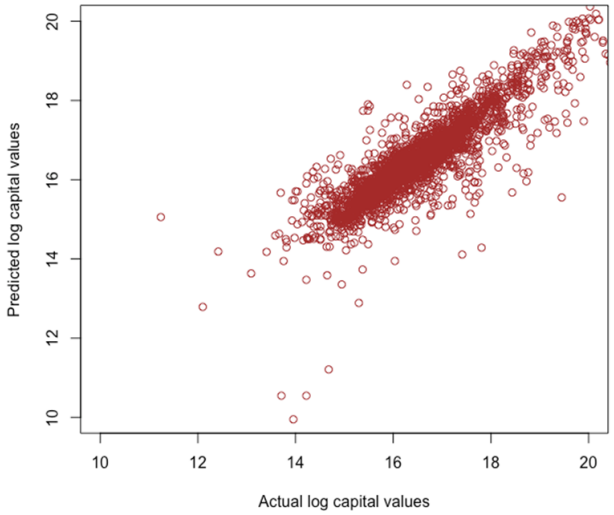
For the current context, property value is a function of land area (L), built area (B), residential use dummy (R), and a vector of cluster dummies (C) such that:

$$V = f(L, B, R, C)$$

To predict V, random forest model was set up where 75% of the data was used to predicted log of V using the logs of L, B, R and C. The remaining 25% of the sample was reserved for cross-validation, a technique used to assess the model's predictive performance on unseen data, thus providing insights into its generalizability.

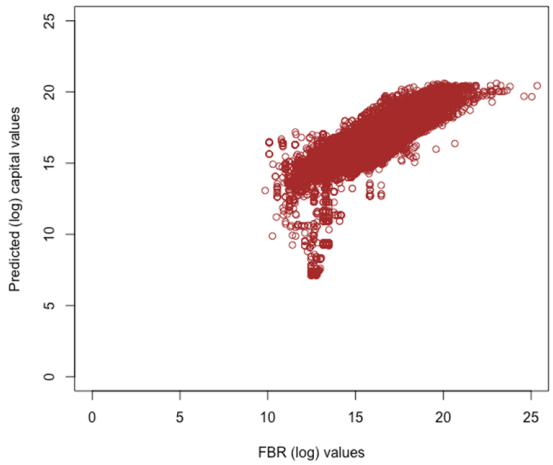
Figures E.4 and E.7 compare predicted and actual capital values in the cross-validation sample, in logs and levels, respectively.

Figure E.4: Relationship between predicted and actual log capital values



The entire valuation sample was then used to train the random forest model and predictions were made for the full valuation sampling frame where we had FBR values. Figure E.5 shows that the correlation between predicted values and FBR values was positive but not as strong as with the cross-validation sample in Figure E.4.

Figure E.5: Relationship between predicted and FBR capital values for the valuation sampling frame

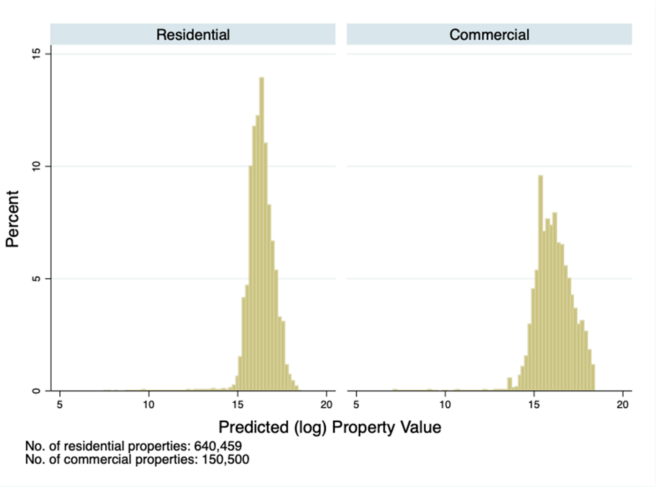


Source: IDEAS-LUMS Property Valuation Survey 2023

The final step in this process was to set up a random forest model and predict values for all residential and commercial properties in the cadastre. This was done by fixing the number of trees to 100 in the final specification and the number of variables used at each split to 2.

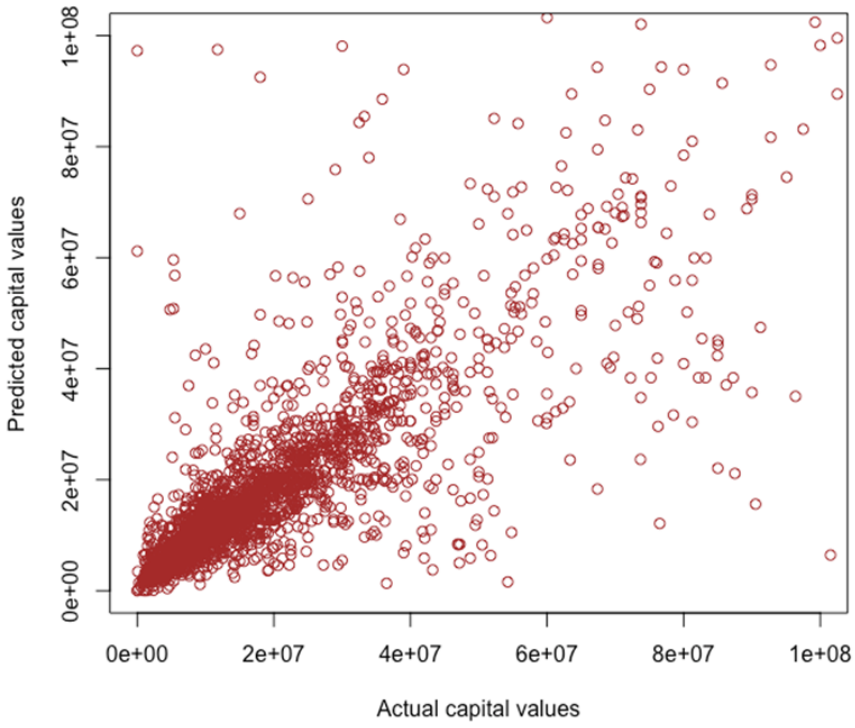
As expected, both residential and commercial property value distributions are right-skewed with few very highly valued properties (see Figure E.6).

Figure E.6: Distributions of predicted property values by property use



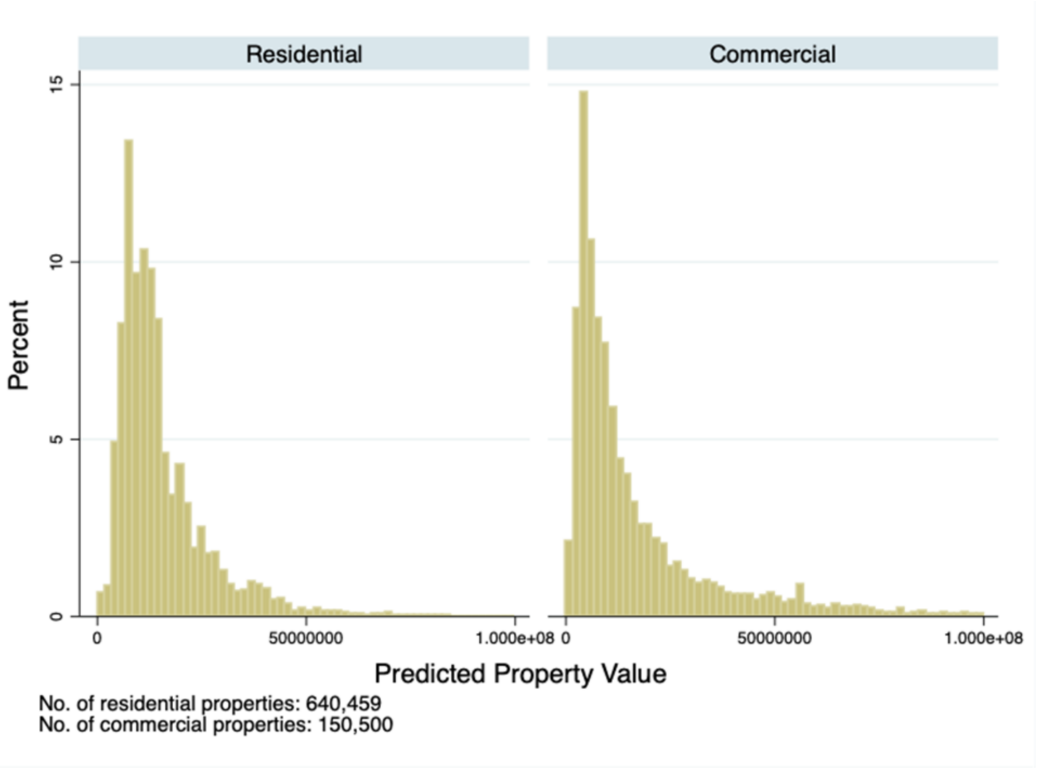
Source: IDEAS-LUMS Property Valuation Survey 2023
Notes: Values restricted to PKR 800 million for better visualization

Figure E.7: Relationship between predicted and actual capital values for the cross-validation sample



Source: IDEAS-LUMS Property Valuation Survey 2023

Figure E.8: Distributions of predicted (log) property values by property use



Source: IDEAS-LUMS Property Valuation Survey 2023
Notes: Values restricted to PKR 800 million for better visualization

F Survey Data Details

F.1 Frontline Tax Officers Preference Elicitation

Figure F.1: Preference Elicitation Dashboard



Source: IDEAS-LUMS Local Tax Officials and Political Workers Survey 2025
Notes: Figure shows screenshot from Android Application developed for eliciting respondents' preferred tax schedule. Respondents were shown property characteristics along with a scale with a default value at Lahore's average tax rate of 0.04%. Respondents were then asked whether they preferred a higher or a lower tax rate than Lahore's average on this property and they the scale was adjusted based on respondents response and their perceptions about tax compliance (Panel A). Once all 6 properties were asked, the respondent was shown a summary table and the shape of their preferred tax schedule (blue line) overlaid with tax schedules of the type of treatment group they belonged to e.g. in the current vignette, the respondent was assigned to see comparsion of tax schedules of Medium-to-High valued property owners (red line) against Govt. sanctioned Jan 2025 tax schedule (green line) (Panel B). Respondents were then given a chance to revise their preferred tax rate the updated graph was shown to them as well (Panel C).

F.2 Frontline Tax Officers Conjoint Experiment

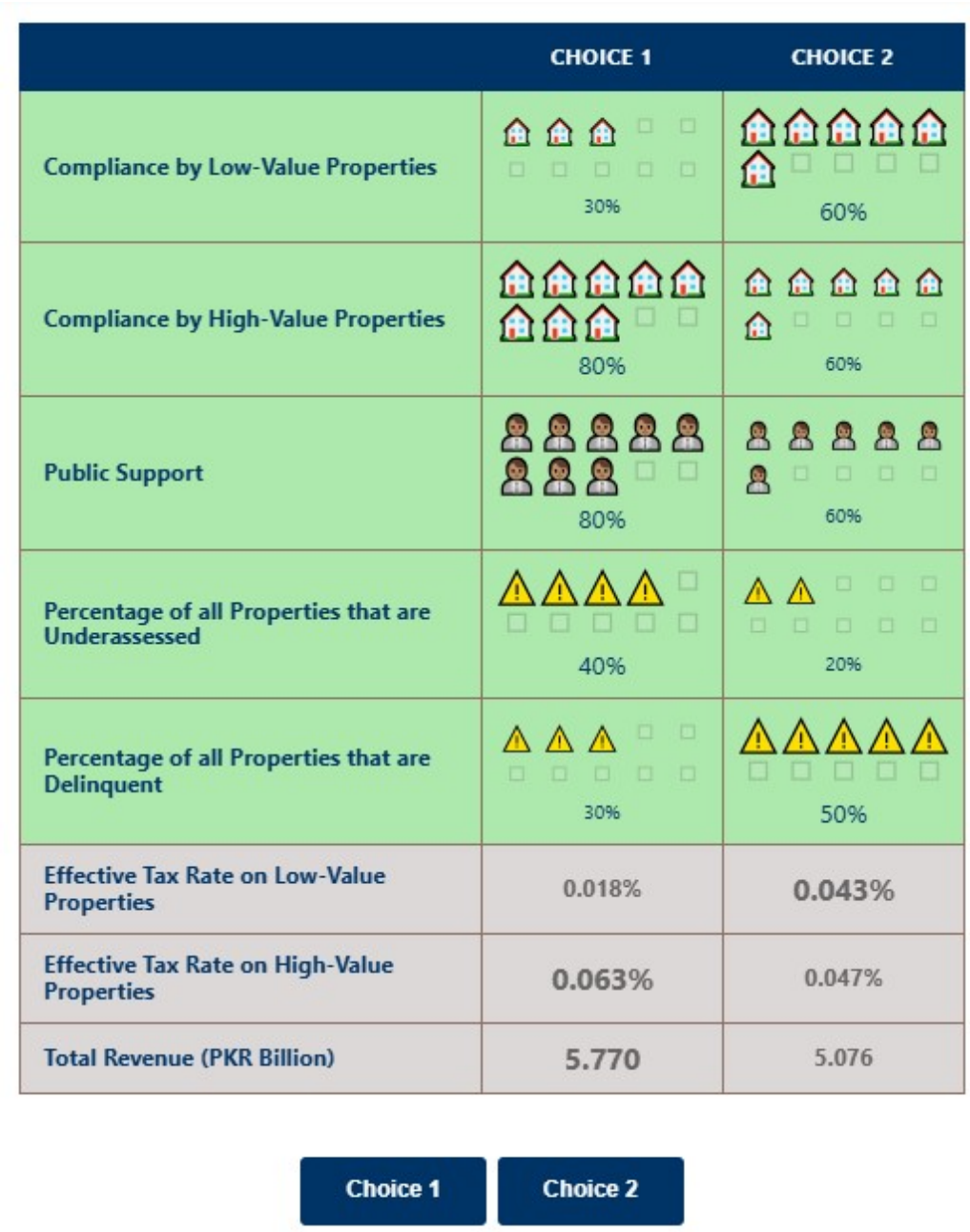


Figure F.3: Example Choice Screen from the Officers Survey Conjoint Experiment

Notes: The figure shows a pair of scenarios presented to tax officials. Respondents select the scenario they consider most likely to continue functioning effectively over the next 3–5 years. The five attributes shown in green are compliance by low-value properties, compliance by high-value properties, public support, the percentage of all properties that are underassessed, and the percentage that are delinquent. The rows below report the effective tax rates on low- and high-value properties and total revenue in PKR billion. The values shown illustrate one choice screen.

F.3 Taxpayers Conjoint Experiment

F.3.1 Taxpayers' survey: sampling and recruitment

The taxpayer survey draws on an existing sample of properties with independent remote-sensing measurements. These measurements provide independent information about respondents' own properties. Sampling within this frame therefore links the survey experiment, demand for property evidence and the administrative analysis to the same properties.

The remote-sensing sample comprises 10,564 properties across 48 of Lahore's Excise and Taxation circles, selected from an administrative register of approximately 600,000 properties. Using ESRI satellite imagery from May 2026, the research team independently delineated parcel boundaries, calculated land area from the resulting polygons and predicted covered area using a machine-learning model. These measurements link to administrative records through Property Identification Numbers. They provide a cross-sectional benchmark for the same identified properties, rather than annual measurements of their physical characteristics.

The measurement frame is spatially broad but selective on property type because dense, contiguous and irregular building layouts constrain parcel delineation (Figure A.6). Table F.1 reports the resulting coverage differences. Relative to non-selected properties, the remote-sensing sample contains larger plots, greater covered area and higher liabilities; normalised differences for these characteristics range from 0.20 to 0.32. Comparisons with non-selected organised or gated properties are closer, with normalised differences across the reported characteristics of 0.08–0.14. These comparisons describe observable differences in coverage; the survey findings pertain to participating taxpayers within the imagery-covered frame, rather than all Lahore taxpayers.

Table F.1: Remote-sensing sample and comparison properties

	RS selected	Organised/ gated	Non-selected	Normalised difference vs org./ gated	Normalised difference vs non-selected
Land area (sq. yd)	392.7	276.3	179.4	0.14	0.25
Covered area (sq. ft)	3,497	2,723	1,736	0.09	0.20
Tax liability (Rs)	9,670	7,986	4,184	0.09	0.32
Population density	118.6	117.3	114.8	0.08	0.21
Properties	10,564	156,983	546,108		

Notes: Group means from the project comparison. Organised/gated properties are non-selected properties recorded as organised, planned or gated; they are a subset of all non-selected properties. The normalised difference is $(\bar{x}_{\text{RS}} - \bar{x}_g)/\sqrt{(s^2_{\text{RS}} + s^2_g)/2}$. Density is reported in the source units. The comparison extract and the historical sampling frame differ in coverage.

Using this sampling frame, for the taxpayer survey, interview targets were specified by tax circle, locality and administrative land-area category. Eligible properties were grouped into four land-area bands: 1–5 marla, above 5 through 10 marla, above 10 through 20 marla, and above 20 through 40 marla. The overall target of 2,100 interviews was allocated across localities in proportion to their aggregate recorded covered area. Within each locality, targets were apportioned across land-area-by-storey cells in proportion to their eligible property counts. Properties were randomly ordered within each cell to construct primary and reserve lists. Out of the total

sampling frame of 10,564 properties, fieldwork yielded 2,100 completed interviews from 6,088 approached properties. Table F.2 summarizes the survey sample.

Table F.2: Sample definitions and observation flow

Stage or analysis	Count	Definition
Remote-sensing sampling frame	10,564	48 circles
Approached properties	6,088	46 observed circles; unique property identifiers
Completed interviews	2,100	45 observed circles
Refusals	3,988	All other recorded survey statuses
Not approached	4,476	Frame less approached properties
Substantive information-offer answers	2,099	One non-substantive answer

Property counts refer to unique properties; the experiment has one completed owner interview per property. Frame coverage, survey completion and validity of a particular measurement are different denominators. Valid imagery measurements are reported separately for each attribute and linked analysis. The broader administrative panel contains 23,233 usable property identifiers; all 6,088 approached properties link to it. Earlier baseline citizen and officer surveys are separate sources.

Table F.2 compares completed interviews, approached properties without a completed interview, and properties not approached on observed administrative characteristics. It shows that citizens who completed the surveys are statistically similar to citizens who refused or were not surveyed.

F.3.2 Conjoint Exhibit



Figure F.4: Example Choice Screen from the Taxpayers Conjoint Experiment

Notes: The figure shows an example choice screen from the citizen conjoint experiment. Respondents choose between two assessment situations for their own property. Each option displays the assessment method, any measurement error, the resulting bill, any offered reduction and informal payment, the final recorded bill, and total out-of-pocket payment. In this example, Option A involves an inspector and an informal payment in exchange for a lower recorded bill, while Option B involves automated satellite assessment without a payment or negotiated reduction.

G Additional Results

G.1 Recording around the five-marla cutoff

The diagnostics use the same properties, category labels, recorded-land outcome and two-marla estimation bandwidth within each comparison. Recorded land comes from the MIS 2021–22

extract dated 9 February 2022; the RS measurements are later independent reference measurements. The analysis categories are held fixed across specifications. Within the estimation window, two properties differ between the retained A/non-A classification and the historical MIS classification, one in each group. The comparison is therefore not a reconstruction of historical category membership.

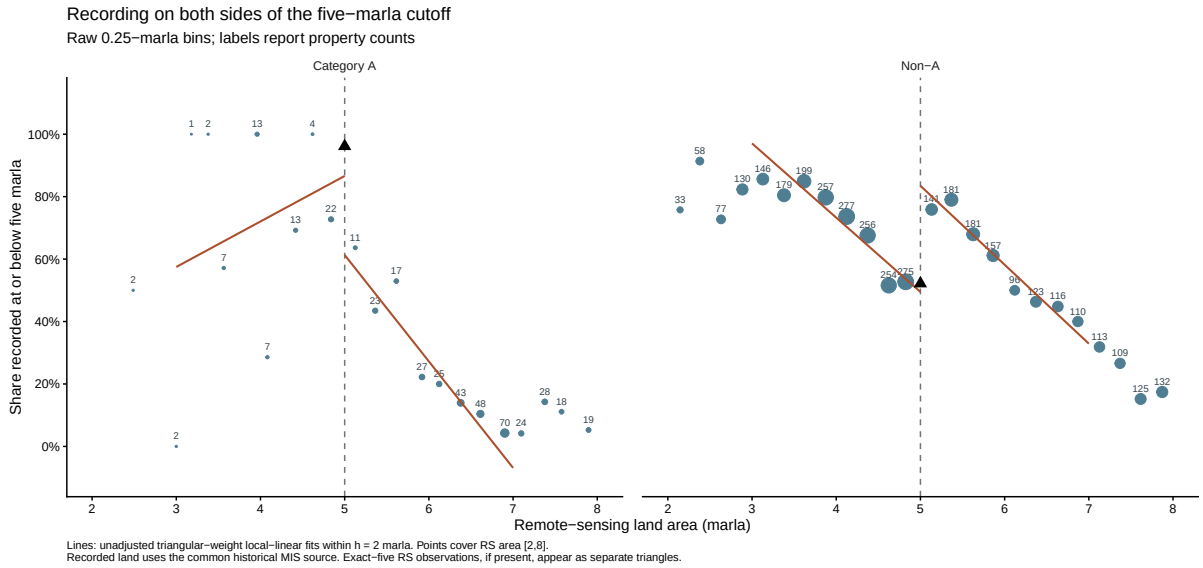


Figure G.1: Recording on both sides of five marla. The panels show raw recording proportions over RS land area $[2, 8]$ and conventional unadjusted local-linear fits within the two-marla estimation window. Triangles identify observations with RS area exactly five. Table G.1 reports the fitted levels and jumps.

Cutoff levels and bias correction. Table G.1 reports the fitted probability on each side of five and both conventional and bias-corrected jumps. With the main-road control evaluated at the same reference for both groups, the non-A fitted probability rises from 0.514 to 0.820, giving a conventional jump of 0.305. Bias correction increases the jump to 0.337. Without controls, the conventional non-A jump is 0.340. Thus, the positive fitted jump is not created by the quadratic bias correction or by the covered-area control. It is algebraically feasible because the fitted probability below the cutoff is substantially below one. Category A's conventional main-road-only jump is −0.287, compared with the bias-corrected estimate of −0.223. The conventional difference between the groups' jumps is 0.592; the bias-corrected difference is 0.560.

Table G.1: Fitted cutoff levels and conventional recording discontinuities

Control	Group	Left	Right	Conventional	BC jump
None	Category A	0.866	0.613	-0.253	-0.176
None	Non-A	0.495	0.835	0.340	0.372
Main road	Category A	0.871	0.584	-0.287	-0.223
Main road	Non-A	0.514	0.820	0.305	0.337

The FY2026–27 benchmark with both controls also has a positive conventional non-A jump (0.279), compared with 0.307 after bias correction (Table G.3). For the historical outcome, the fixed complete-case comparison contains 359 Category-A and 2,994 non-A properties within

the estimation window. The conventional difference between jumps is 0.592 with main-road status and 0.570 with both controls; the corresponding bias-corrected differences are 0.560 and 0.536. The main-road-only baseline avoids conditioning on the potentially endogenous predicted covered-area measure.

Raw recording probabilities and RS measurement. The raw non-A proportions in $[4.75, 5)$ and $(5, 5.25]$ are 52.7% and 75.9%. In the closer intervals $[4.9, 5)$ and $(5, 5.1]$, they are 80.3% and 81.6%. These are descriptive averages, not estimates of the limiting probabilities or tests that the discontinuity is zero. The lower-side pattern explains why the positive fitted jump can coexist with a binary outcome, but it also makes the interpretation depend on the quality and distribution of the RS measurements.

The RS layer contains a cluster near 4.7693 marla, identified as a geometry artefact in the concentration analysis. The discontinuity sample includes 125 non-A observations within 0.001 marla of that value; 95 are recorded above five. They account for 95 of the 130 recorded-above-five cases in $[4.75, 5)$. Among the remaining 150 observations in that interval, 76.7% are recorded at or below five, compared with 75.9% in $(5, 5.25]$. The analogous FY2026–27 proportions are 76.0% and 74.5%. This decomposition describes the raw observations; it is not a discontinuity re-estimated after excluding the flagged values. The reported RD estimates have not been subjected to that exclusion. A disagreement with RS does not alone establish over-recording relative to historical true land area.

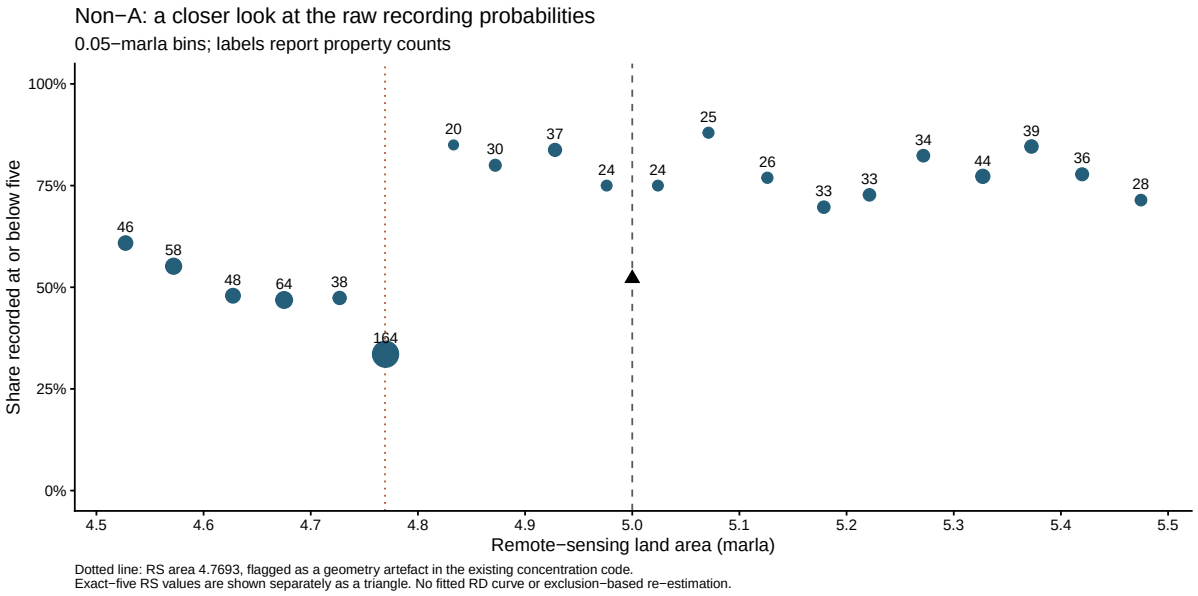


Figure G.2: Non-A recording probabilities close to five marla. The dotted line marks the RS-value cluster near 4.7693; the dashed line marks five. The triangle reports the exact-five RS observations. The figure describes the raw observations and does not report an exclusion-based RD estimate.

Assessment-circle support and exact-five observations. Table G.2 gives the independent assessment-circle support on each side. The 95 Category-A lower-side properties span 13 circles. Twenty-six of those properties have RS area exactly five, span five circles, and include 25 properties recorded at or below five. The 46 exact-five non-A properties span 21 circles, with a recorded-at-or-below-five share of 52.2%. The small numerical shift of the computational

cutoff implements the lower-side convention for exact-five observations. It does not remove their influence. The results reported here do not include an exact-five exclusion sensitivity or a leave-one-circle-out analysis.

Table G.2: Property and assessment-circle support at the five-marla cutoff

Group	N_-	N_+	Circles −	Circles +	Exact-five N
Category A	95	264	13	23	26
Non-A	1,889	1,105	56	43	46

The two-sided evidence answers whether the positive non-A jump is present before bias correction. It does not by itself establish a causal exemption effect. That interpretation additionally requires that other recording determinants, benchmark measurement differences and changes in sample composition do not generate a differential break at five. The later RS measurement is not a contemporaneous observation of historical true land area. The concentration evidence and the fitted recording contrast are therefore interpreted together, with the measurement and support issues above remaining relevant to identification.

G.2 FY2026–27 benchmark and bandwidth sensitivity

Table G.3: FY2026–27 benchmark: bandwidth and covered-area control sensitivity

Controls	h	$\widehat{d_5}$	Boot s.e.	95% interval	N_A	N_{nonA}
Both	1.0	0.666	0.232	[0.266, 1.155]	74	1,768
Both	1.5	0.702	0.222	[0.143, 1.006]	121	2,443
Both	2.0	0.704	0.219	[0.129, 0.974]	208	2,994
Both	2.5	0.720	0.205	[0.170, 0.958]	242	3,423
Both	3.0	0.749	0.192	[0.226, 0.964]	260	3,771
Main road	1.0	0.632	0.206	[0.199, 0.961]	74	1,768
Main road	1.5	0.674	0.237	[0.113, 1.016]	121	2,443
Main road	2.0	0.667	0.241	[0.101, 0.971]	208	2,994
Main road	2.5	0.689	0.226	[0.121, 0.949]	242	3,423
Main road	3.0	0.719	0.214	[0.170, 0.958]	260	3,771

Notes: The outcome in this benchmark is FY2026–27 recorded land area at or below five; the property sample and data period differ from the historical baseline in Table 6.1. Bandwidths are in marla. Both controls means main-road status and log RS covered area. Eligibility and category-consistency screens are fixed within this table. The same circle-bootstrap draws are used across specifications: 999 paired draws succeed at every reported bandwidth. Extreme successful draws are retained. The bias-to-estimation bandwidth ratio is 1.6521.

The one-marla specification has a bootstrap standard error of 0.232 with both controls and 0.206 with main-road status only. All 999 paired resamples return estimates at each reported bandwidth. Bootstrap standard errors across the reported specifications range from 0.192 to 0.241. Numerical completion does not establish robustness to influential circles; no leave-one-circle-out assessment is reported here.

G.3 Robustness of compensating revenue equivalents

Sample definition and denominator screens. The full sample contains 289 officers. The group-specific calculations retain 212 officers for low-value collections and 221 for higher-value collections, excluding 77 (26.6%) and 68 (23.5%), respectively. Requiring both collection coefficients to be positive retains 178 officers and excludes 111 (38.4%). In implementation, positivity means a posterior mean above 10^{-10}. These restrictions define the officers for whom additional collections have the compensating interpretation in equation (**??**). A negative collection coefficient reverses that inequality, while a zero coefficient cannot offset the exchange through that margin. Non-positive estimates can reflect substantive judgements about the displayed outcomes or estimation uncertainty; the saved summaries do not distinguish these explanations. The 60.2% recovery-priority result uses all 289 officers and does not impose these restrictions.

The stronger screen requires $\hat{\beta}_{kj} - 1.96 s_{kj} > 0$, where s_{kj} is the marginal posterior standard deviation of the collection coefficient. It retains 87 officers (30.1%) for low-value collections and 97 (33.6%) for higher-value collections. This is a normal-approximation screen relative to each coefficient's uncertainty, rather than a common absolute coefficient cutoff or a directly calculated posterior sign probability.

Table G.4: Sampling uncertainty in median revenue thresholds

Sample / collection group	N	Median	Boot s.e.	95% interval
Group-specific / Low	212	4.96	1.15	3.62 to 7.68
Group-specific / High	221	13.85	2.87	8.95 to 21.28
Same officers / Low	178	4.85	1.28	3.62 to 7.87
Same officers / High	178	15.96	3.83	10.52 to 23.63
Stronger coefficient screen / Low	87	4.24	1.13	0.98 to 6.52
Stronger coefficient screen / High	97	16.66	3.36	9.13 to 22.20

Notes: Amounts are PKR millions for one percentage point less underassessment and one point more delinquency. Group-specific samples require the relevant posterior-mean collection coefficient to be positive; the common sample requires both to be positive. The stronger screen requires $\hat{\beta}_{kj} - 1.96 s_{kj} > 0$, where s_{kj} is its marginal posterior standard deviation. Intervals resample officers 1,999 times within each stated subsample, holding their saved coefficient estimates fixed.

Stability of the conditional medians. The low-value median is PKR 4.96 million in the group-specific sample, PKR 4.85 million among the common officers and PKR 4.24 million under the stronger screen. The corresponding higher-value medians are PKR 13.85 million, PKR 15.96 million and PKR 16.66 million. All remain positive, and their reported bootstrap intervals exclude zero. These comparisons support the stability of the positive median trade-off within the stated samples; they do not recover a median for all officers. Because $\nu_k > 0$ and the retained $\hat{\beta}_{kj} > 0$, the share with positive $\widehat{MRS}_{kj}$ is exactly the share with $\hat{\omega}_j > 0$. It is the same sign statistic expressed in revenue units.

Posterior summaries and uncertainty. The officer estimates include posterior means, medians and marginal standard deviations. The reported revenue equivalents divide posterior means, and the stronger screen uses the marginal standard deviations. Table G.4 resamples officers within each retained sample, holding their coefficients and sample eligibility fixed. It measures variation in these conditional medians under officer resampling; it does not propagate the poste-

rior uncertainty in the coefficients or dependence arising from shared staffing. Joint coefficient draws or cross-coefficient covariances are not included in the terminal-summary file used here. Marginal summaries alone do not determine the posterior distribution of a coefficient ratio, so these bootstrap intervals are not posterior credible intervals for the MRS. Using medians and the stronger screen limits sensitivity to small denominators without resolving that distinction.

G.4 Field-based ranking and alternative inputs

Table G.5: Fiscal comparisons using a field-based value ranking

Inputs replaced	Net change (PKR)	Change (%)	ΔK	95% interval for ΔK
RS land only	−872	−5.35	0.0388	[−0.0023, 0.0840]
RS covered only	423	2.59	−0.0008	[−0.0215, 0.0165]
Both RS inputs	−473	−2.90	0.0376	[−0.0096, 0.0856]
Field land only	−466	−2.86	0.0655	[0.0337, 0.1042]
Both field inputs	872	5.35	0.0490	[0.0117, 0.0919]

Notes: All scenarios use 2,407 properties in 55 circles, the same inherited weights, administrative baseline bills and field formula-value ranking. Baseline mean liability is PKR 16,286.39. Intervals use 999 circle resamples. Joint field replacement changes mean liability by PKR 871.78 [PKR −435.89, PKR 2,282.85]; joint RS replacement changes it by PKR −472.82 [PKR −2, 104.33, PKR 1,021.32].

G.5 Persistence and the destination of changed records

In the retained baseline-to-endpoint sample, 6,928 of 7,143 properties have identical recorded land area in FY2024–25 and FY2026–27. Of the 215 changed endpoints, 94 move closer to the fixed RS land reference, including two that finish within approximately 1%; 121 move farther away. This endpoint comparison complements the five-year persistence results by distinguishing an edit from convergence to the independent reference.

Table G.6: Recorded land changes and convergence to the independent reference

Destination of changed land record	Properties
Moves closer and ends within approximately 1% of RS	2
Moves closer but remains outside that range	92
Moves farther from RS	121
All changed land records	215

Notes: Baseline is FY2024–25 and endpoint FY2026–27. Closer or farther requires a change exceeding 10^{-8} in absolute log distance to the fixed RS reference. Agreement requires endpoint absolute log distance no greater than 0.01. The sample retains the 7,143 properties used for this existing endpoint comparison.

G.6 Verification targeting across inspection shares and error tolerances

Table G.7: Verification targeting: land area

	Share of properties inspected				
	5%	10%	20%	30%	50%
Error threshold $\lvert\Delta \log X\rvert > 0.10$	*base rate 65.8%*				
Precision (%)	79.9	82.8	84.4	86.5	83.6
Gain over random inspection (pp)	14.0***	17.0***	18.6***	20.7***	17.8***
standard error	(3.6)	(2.2)	(2.5)	(2.0)	(1.7)
Recall (%)	6.0	12.6	25.7	39.4	63.5
Absolute error captured (%)	22.9	39.2	57.6	70.5	84.8
Error threshold $\lvert\Delta \log X\rvert > 0.25$	*base rate 39.1%*				
Precision (%)	69.1	76.3	79.4	80.7	70.7
Gain over random inspection (pp)	30.0***	37.3***	40.3***	41.6***	31.7***
standard error	(7.0)	(4.4)	(4.0)	(2.9)	(1.5)
Recall (%)	8.8	19.5	40.6	61.8	90.5
Absolute error captured (%)	22.9	39.2	57.6	70.5	84.8
Error threshold $\lvert\Delta \log X\rvert > 0.50$	*base rate 24.6%*				
Precision (%)	66.2	74.6	75.7	72.1	46.8
Gain over random inspection (pp)	41.6***	49.9***	51.1***	47.5***	22.2***
standard error	(7.0)	(4.2)	(4.0)	(2.7)	(2.4)
Recall (%)	13.4	30.2	61.5	87.8	95.1
Absolute error captured (%)	22.9	39.2	57.6	70.5	84.8
Properties inspected	139	279	559	838	1,398

Properties are ranked by the discrepancy score $Q_i = |\log X_i^{RS} - \log X_i^{admin}|$ and the highest-scoring share is inspected. The site measurement never enters the score; it appears only in the outcome.

A property is a site-confirmed error when the administrative record differs from the field measurement by more than the stated tolerance in absolute log points. Precision is the share of the inspected list that is a confirmed error. Under random inspection precision equals the base rate, which is reported in each panel heading, so the gain row is the operationally meaningful quantity.

Recall is the share of all confirmed errors that the list finds. Absolute error captured is the share of the sample's total $|\Delta \log X|$ that sits on the inspected list, which weights a large error more heavily than a marginal one.

Standard errors from a circle cluster bootstrap, 1,000 replications, seed 20260825, resampling the 60 tax circles in the validation sample. Stars from the bootstrap distribution of the gain: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The score is a fixed function of the two administrative-side sources with no parameter estimated on the validation data, so no cross-fitting is required. Any tuned score would need circles partitioned before evaluation.

The error is defined on the recorded attribute rather than on a formula value, following the chapter's own rule that outcomes are defined on attributes because the statutory schedule is not recoverable for the pre-reform period.

Table G.8: Verification targeting: covered area

	Share of properties inspected				
	5%	10%	20%	30%	50%
Error threshold $\lvert\Delta \log X\rvert > 0.10$	*base rate 72.8%*				
Precision (%)	92.5	88.5	88.1	88.5	84.2
Gain over random inspection (pp)	19.7***	15.7***	15.3***	15.7***	11.4***
standard error	(3.1)	(2.2)	(2.2)	(1.9)	(1.3)
Recall (%)	6.3	12.1	24.2	36.4	57.8
Absolute error captured (%)	14.7	23.7	39.2	51.6	69.6
Error threshold $\lvert\Delta \log X\rvert > 0.25$	*base rate 44.8%*				
Precision (%)	76.9	74.3	75.1	72.0	64.5
Gain over random inspection (pp)	32.1***	29.6***	30.3***	27.2***	19.7***
standard error	(5.2)	(3.5)	(3.4)	(2.3)	(1.5)
Recall (%)	8.5	16.6	33.5	48.2	72.0
Absolute error captured (%)	14.7	23.7	39.2	51.6	69.6
Error threshold $\lvert\Delta \log X\rvert > 0.50$	*base rate 20.2%*				
Precision (%)	64.9	62.5	55.2	48.3	33.0
Gain over random inspection (pp)	44.7***	42.3***	35.0***	28.1***	12.8***
standard error	(7.2)	(4.8)	(3.1)	(2.6)	(1.4)
Recall (%)	16.0	30.9	54.7	71.8	81.8
Absolute error captured (%)	14.7	23.7	39.2	51.6	69.6
Properties inspected	134	269	538	807	1,345

Properties are ranked by the discrepancy score $Q_i = |\log X_i^{RS} - \log X_i^{admin}|$ and the highest-scoring share is inspected. The site measurement never enters the score; it appears only in the outcome.

A property is a site-confirmed error when the administrative record differs from the field measurement by more than the stated tolerance in absolute log points. Precision is the share of the inspected list that is a confirmed error. Under random inspection precision equals the base rate, which is reported in each panel heading, so the gain row is the operationally meaningful quantity.

Recall is the share of all confirmed errors that the list finds. Absolute error captured is the share of the sample's total $|\Delta \log X|$ that sits on the inspected list, which weights a large error more heavily than a marginal one.

Standard errors from a circle cluster bootstrap, 1,000 replications, seed 20260825, resampling the 58 tax circles in the validation sample. Stars from the bootstrap distribution of the gain: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The score is a fixed function of the two administrative-side sources with no parameter estimated on the validation data, so no cross-fitting is required. Any tuned score would need circles partitioned before evaluation.

The error is defined on the recorded attribute rather than on a formula value, following the chapter's own rule that outcomes are defined on attributes because the statutory schedule is not recoverable for the pre-reform period.